

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (VPA DD)

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VERSION **v.2.3**

RELATED SUPPORT

- [Programme of Activity requirements](#)

- [TEMPLATE GUIDE VPA Design Document](#)

This document contains the following sections

Section A – Description of project

Section B – Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

Section D – Summary of Safeguarding Principles and Gender Sensitive Assessment

Section E – Summary of Local stakeholder consultation

Section F – Eligibility and inclusion criteria for VPAs inclusion

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2- Contact information of VPA Implementer (mandatory)

KEY PROJECT INFORMATION

Type of VPA	☐ Real case VPA ☒ Regular VPA
Scale of VPA Note that a VPA can be of one scale. Please select applicable scale accordingly.	☐ Microscale ☐ Small scale ☒ Large scale
Title of corresponding real case VPA (if applicable)	PoA GS11139- Access to Improved Cooking Solutions Programme- Efficient Kitchens in India-VPA01
GS ID of real case VPA (if applicable)	GS11358
GS ID of VPA	GS12533
Title of VPA	PoA GS11139- Access to Improved Cooking Solutions Programme- Efficient Kitchens in Gujarat, Uttar Pradesh, and Madhya Pradesh - VPA02
Time of First Submission Date	09/01/2024
Date of Design Certification	06/05/2024
Version number of the VPA-DD	03
Completion date of version	29/05/2024
Coordinating/managing entity	Climate Impact Partners Limited
VPA Implementer (s)	Greenway Grameen Infra Private Limited
Project Participants and any communities involved	SDG 13 VENTURES PTE. LTD
Host Country (ies)	India
GS ID and Title of applicable Design Certified VPA	PoA GS11139- Access to Improved Cooking Solutions Programme- Efficient Kitchens in India-VPA01 (GS11358)
GS ID and Title of applicable Performance Certified VPA	PoA GS11139- Access to Improved Cooking Solutions Programme- Efficient Kitchens in India-VPA01 (GS11358)
Activity Requirements applied	☒ Community Services Activities

	☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Other Requirements applied	-Programme of Activity Requirements and Procedures, version 2.1 -Tool 30: Calculation of the fraction of non-renewable biomass, version 4.0” -GHG Emissions Reduction & Sequestration Product Requirements, Version 2.2 -Community Services Activity Requirements, Version 1.2 -Stakeholder Consultation and Engagement Requirements, Version 2.1 -Stakeholder Consultation and Engagement Guidelines, Version 2.1 -GS4GG Principles & Requirements, Version 1.2 -Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 4.0 -Usage Survey Guidelines, version 2.0. -SDG Impact tool, version 1.3
Methodology (ies) applied and version number	Technologies and Practices to Displace Decentralized Thermal Energy Consumption, Version 4.0
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
VPA Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

SUSTAINABLE DEVELOPMENT GOALS TARGETED	SDG IMPACT (DEFINED IN B.6.)	ESTIMATED ANNUAL AVERAGE	UNITS OR PRODUCTS
13 Climate Action (mandatory)	Emission reduction	39,815	VERs
1. No Poverty	Estimated monetary savings computed from fuel saved.	27.32	Rupee/household /day
5. Gender Equality	Average time saved by women for fuel collection	1.07	Hours/household /day
7. Access to clean energy	Number of households with access to basic service provided by the project device	20,000	Number
8. Decent Work and economic growth	Total number of jobs split by gender	50 (30 male and 20 female)	Number of people

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

>> The purpose of this project activity is to promote and support distribution of high quality and efficient fuelwood cooking stoves to domestic users in the Indian states of Gujarat, Uttar Pradesh, and Madhya Pradesh where the VPA will be located. The proposed VPA will be in the three states, which falls within the boundary of the registered Programme of Activities under which this proposed VPA is being included.

The voluntary project activity has been developed by Climate Impact Partners Limited and will be implemented by Greenway Grameen Infra Private Limited, who manufactures the stoves. The project activity will promote the Greenway Jumbo improved firewood stove, which has a rated thermal efficiency of 38%.

The project activity targets to distribute 20,000 Greenway Jumbo stoves which will displace the use of inefficient baseline cooking technologies that consume more non-renewable biomass. The stoves targets to benefit end-users who in the baseline were using inefficient biomass stove technologies. From the applied methodology (Technologies and Practices to Displace Decentralized Thermal Energy Consumption, Version 4.0), the baseline scenario has been determined as the consumption of non-renewable biomass using inefficient cooking devices to deliver thermal energy among household end-users.

The project activity has been designed with an incentive scheme that entails subsidising the cost of the stove. The Greenway Jumbo stove is priced at INR 3499. For this project activity, the stove is being distributed to the community at a subsidised price of INR 400. Carbon financed is being utilized to support the uptake of the stoves by households. Such a facility is not available to the baseline technology users. This is an incentive mechanism which is being utilised to enable uptake of the project stoves and also to discourage the use of the baseline stoves.

Goal and Objectives of the Project Activity

The goal of the project activity is to promote use of energy efficient fuelwood cookstoves which will significantly reduce use on non-renewable biomass for cooking leading to fuel savings. Use of the project technology will displace GHG emissions from the thermal energy consumption of households and contribute towards reduced household air pollution.

Specific Objectives

1. To increase access to improved fuelwood cookstoves within the project area which will replace traditional inefficient biomass cookstoves used in the baseline.
2. To contribute towards improved health among end-users through the reduction of household air pollution.
3. To reduce greenhouse gas emissions associated with cooking with non-renewable biomass in the targeted populations.

The location of the project activity

The project activity will be located in the states of Gujarat, Uttar Pradesh, and Madhya Pradesh, within the Republic of India. India lies within Latitudes 8° 4' and 37° 6' north, and longitudes 68° 7' and 97° 25' east.

Duration

The project start date is 16/10/2023 which marks the date when the first stove was sold under the project activity. The crediting period start date is 16/10/2023. The total length of crediting period is 15 years (5 years renewable twice).

Registration with other Carbon standards

The project activity is not registered with any other carbon standard, nor is it seeking registration under any other GHG programs, and hence there is no likelihood that double counting would occur. Additionally, each cookstove shall be assigned a unique serial number which gives a distinction on the end-users which will further ensure there is no double counting.

Carbon credits ownership

The legal ownership of the products generated belongs to Climate Impact Partners Limited (CME). The end-users' consent to the transfer of carbon credits generated from using the project technology at the time of sale to the project developer (Climate Impact Partners Limited).

Implementation plan

Greenway Grameen Infra Private Limited is the project implementor and will be directly responsible for production of the stoves, distribution and maintaining the sales database. They will also be responsible for collecting baseline information, co-ordinating the annual monitoring process and maintenance of the stoves.

The project activity's projection is to distribute 20,000 improved cookstoves which will be installed within end-user's premises located within the target area. The expected annual average emission reductions are 39,815tCO₂e, and the total emission reductions during the first crediting period are 199,078 tCO₂e.

Host Country regulations on emission reductions

The host country has not set any regulations setting a cap on the emission reduction generated by projects. There is also no possibility for the host country to trade emissions that include the scope of the proposed project activity.

A.1.1. Eligibility of the VPA under approved PoA

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Table 2 Eligibility for VPA inclusion as per PoA requirements

NO.	ELIGIBILITY CRITERION	DESCRIPTION/ REQUIRED CONDITION	DESCRIPTION OF THE VPA IN RELATION TO THE CRITERIA, MEANS OF VERIFICATION AND SUPPORTING EVIDENCE FOR INCLUSION
1.	<i>Geographical boundaries of V/CPAs must be consistent with the geographical boundary of the PoA;</i>	The VPA shall be located within the PoA boundary of India	The VPA is being implemented in the states of Gujarat, Madhya Pradesh and Uttar Pradesh which are within the Republic of India (the PoA boundary).
2.	<i>Conditions to avoid double counting of Impacts, such as unique identifications of product and end user locations (e.g. programme logo);</i>	Identification of each device through unique identification number. Each VPA will show that it is exclusive to the PoA and not registered as another project activity or VPA under another PoA.	Greenway stoves under this project activity have a unique serial number that ensures they are uniquely identifiable to this project. Further, each end user shall have only one registered project stove to ensure there is no double counting. An updated sales database will be maintained and will record the number of stoves sold to each end-user. Only one stove shall be eligible for crediting. The PP has also signed a letter confirming that the VPA is only registered to the PoA.
3.	<i>Conditions to confirm that V/CPAs are neither registered as project activities with other offset Schemes, included in other registered PoAs, nor the project activities that have been deregistered;</i>	Confirmation that the VPA is neither registered with other carbon standards	The CME has declared that the VPA is neither registered as a project activity with GS or any other standard or as a VPA of another PoA. The PP conducted a search on the Gold Standard, CDM and VERRA registries to check the following: (i)

			Registration of the project with other standards (ii) similar projects registered within the same project boundary. (iii) projects using the same project technology as the project and (iv) whether the project activity had been deregistered by any of the standards. The search was conducted on 23/04/2024 and the PP can confirm that the VPA is not registered with other Carbon Standards. However, from the search it was established that there are other programs that were found to be making use of the Greenway stoves as follows: 1. PoA GS10818 - Dissemination of Improved Cookstoves in India by Greenway - Dissemination of Improved Cookstoves in Karnataka by Greenway – (VPA001-VPA026) 2. GS11242- Fair Climate Programme for Sustainable Household Energy (PoA) 3. CDM 9181: MicroEnergy Credits – Microfinance for Clean Energy Product Lines – India (unfccc.int)- POA (CPAs 01-37) To ensure there is no double counting, Greenway have a robust stove identification system
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			where each stove in each project is clearly distinguished by a unique serial number and cannot be included in any other project. The projects are implemented by different project developers and each project maintains a separate and distinct database hence avoiding any risk of double counting. The CME can also confirm that the VPA is not a deregistered VPA which is being introduced again. Although there are other projects whose boundaries overlap with the proposed project, the CME confirms that appropriate procedures have been put in place to distinguish each project database and separate each unique stove serial number being included in their respective project databases and this action prevents double counting other mitigation outcomes (See ICS 6 for more details).
4.	<i>Specification of the technology/measure such as the level and type of service, as well as performance specification based on, inter alia, testing/certification;</i>	The VPA shall involve the distribution and sale of highly efficient biomass burning stoves with 20% efficiency and above. Each VPA will involve the distribution and installation of efficient cookstoves to households and/or communities currently cooking with non-renewable biomass.	As per the lab test results provided by the PP, the efficiency of the Greenway Jumbo stoves was tested and has been confirmed as 38%. The project sells stoves to households only.
5.	<i>Conditions to check the start dates of V/CPAs through documentary evidence;</i>	The start date of the VPA shall not be before the PoA stakeholders' consultation date.	The PoA design consultation was initiated on 14/10/2020. The first stakeholder consultation meeting for the VPA was held

			on 10/11/2023. The VPA start date is 16/10/2023 which marks the date when the first stove was sold under the VPA. The stove database shows the actual distribution dates for the stoves and PP has also submitted supporting carbon waiver signed by the end-user as proof of start date (see Evidence of Start date) and the provided stakeholder's consultation report which shows the date when the consultations were done.
6.	<i>Conditions to ensure compliance with the applicability of the applied methodologies, the applied standardized baselines and the other applied methodological regulatory documents</i>	The VPA shall meet the applicability requirements of the methodology "Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version 4.0" and GS4GG Principles and Requirements.	The VPA applies an approved Gold Standard methodology "Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version 4.0". The applied methodology is applicable to the VPA since the methodology requires that only activities that introduce technologies and/or practices that and non-domestic premises can apply it. Since the project introduces stoves technologies that reduce greenhouse gas (GHG) emissions from the thermal energy consumption of households, the VPA therefore meets this condition.
7.	<i>Conditions to ensure that V/CPAs meet the requirements for demonstration of additionality</i>	Each VPA within the PoA shall demonstrate additionality through the Gold Standard additionality criteria or the latest version of the CDM Methodological Tool for the demonstration and assessment of additionality.	Assessment of additionality has been done at VPA level as outlined in section B.5 of the VPA-DD. Documentation: VPA-DD
8.	<i>Conditions to ensure no diversion of official development assistance;</i>	Confirmation of non-involvement of public	The Project activity have signed and ODA declaration form. The

		funding or ODA from Annex I Parties in VPA	signed form has been made uploaded on SustainCert app.
9.	<i>Target group (e.g. domestic/commercial/industrial, rural/urban, grid connected/offgrid), and where applicable, distribution mechanisms (e.g. direct installation)</i>	Target groups should be identified	The VPA distributes improved fuelwood cooking devices to domestic end users and the distribution mechanism is outlined in section A.1 of the VPA-DD.
10.	<i>Conditions related to sampling requirements for the PoA</i>	Sampling requirements be outlined in section B.2 of the PoA shall be applied for the VPAs included into the programme.	The VPA will apply the latest Guidelines for sampling and surveys for CDM project activities and programmes of activities and the sampling process has been outlined in section B.7.2 of the VPA-DD.
11.	<i>Conditions to confirm that technologies in V/CPAs are eligible</i>	All VPAs to meet eligibility requirements specified in section B.3 of the PoA-DD whereby the stoves utilise non-renewable biomass as feedstock and will use fuelwood as the fuel type	The VPA is installing the improved Greenway Jumbo cookstove whose rated thermal efficiency is 38%. This is above the minimum efficiency of 20% required under the methodology. The details of the stove have been explained in section A.3 of this VPA-DD. The fuel being used is non-renewable biomass feedstock and fuelwood as the only fuel type.
12.	<i>Conditions to be met by each VPA regarding SDG outcomes assessment</i>	Each VPA shall conduct an SDG Outcomes assessment and shall positively contribute to the sustainable development goals identified in the PoA-DD.	An SDG outcomes assessment has been done in the VPA-DD in section B.6
13.	<i>Conditions to be met by each VPA regarding safeguarding principles</i>	Each VPA shall conduct a safeguarding principles assessment in line with the Gold Standard Safeguarding principles assessment requirements.	A safeguarding principles assessment has been done in the VPA-DD as detailed in Appendix 1
14.	<i>Legal Ownership</i>	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated.	End users shall sign a Carbon Waiver Form during purchase which transfers the rights of the carbon credits to the CME (Climate Impact Partners).
15.	<i>On-going financial need</i>	<i>Each VPA to demonstrate on-going financial need</i>	As per para 4.1.52 of Principles and

			requirements, Ongoing Financial Need shall be demonstrated at Design Certification Renewal.
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Eligibility with GS4GG Requirements

The project activity falls under section 3.1.1 of *GS4GG Principles & Requirements* (Version 1.2), section 3 of *GS4GG Community Services Activity Requirements* (Version 1.2) and section 5 of *GS4GG GHG Emissions Reduction & Sequestration Product Requirements* (Version 2.2) with the following eligibility criteria:

Table 3: Demonstration of eligibility with GS4GG requirements

Eligibility Criteria	Justification
(a) Types of Projects Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section. Section 3.1.1 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Pre-identified CSA project types are a) Renewable energy; b) End-use energy efficiency; c) Waste management and handling; d) Water, sanitation, and hygiene (WASH). Section 5.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.2) The Following Project types are eligible for issuance of Gold Standard VERs: a) Renewable Energy Supply; b) End-Use	The project activity involves the distribution of improved cookstoves among household end-users. Therefore, the project activity falls under section 3.1.1 (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end-user of the products and services are clearly identified and when the physical intervention is required at the user end." The project activity falls under section 5.1.1 (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services

Energy Efficiency Improvement; c) Waste Handling & Disposal; d) Land Use and Forests. As per section 4.1.3 of <i>GS4GG Principles & Requirements</i> (Version 1.2), "A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it, or it's referenced in the Gold Standard Product Requirements".	or products, where the end user of the products and services are clearly identified and when the physical intervention is required at the user end." of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.2). The project type is "End-Use Energy Efficiency Improvement" and the Impact Quantification Methodology to be applied is "Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC) v4.0". Method 1 given in the methodology for calculating emission reduction for improved biomass cookstoves is applied.
(b) Location of Project: Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Projects may be located in any part of the world. Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.2) Gold Standard VER Projects may be located in any host country or state. However, where host countries or states have mandatory operational schemes to reduce GHG emissions in any form (e.g.,	The project activity shall be located in the states of Gujarat, Madhya Pradesh and Uttar Pradesh in the Republic of India (refer to section A.2 of the VPA-DD for detailed project area and boundary). There are no mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in India, thus based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.2) is an eligible host country.

cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either: (a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occurs or (b) has committed to retiring eligible units equal to the quantity of Gold Standard VERs. Refer to Annex A of this document.	
(c) Project Area, Project Boundary and Scale: Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. To avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and	The project activity location is the states of Gujarat, Madhya Pradesh and Uttar Pradesh in the Republic of India (refer to section A.2 of the VPA-DD for detailed project area and boundary) and there are no mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in India, thus based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.2) is an eligible host country. Greenway Grameen Infra Private Limited is implementing another Gold Standard Project which uses their project stoves. In addition, the Greenway stoves are also used in another CDM registered project. However, each project maintains a separate and distinct database for each project. Each stove in each project is clearly distinguished by a unique serial number and cannot be included in any other project (see section B.2 and

performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects). Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2): Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. Section 9.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.2) GSVER Projects may be registered as 'large scale', 'small scale' (for the applicability of methodologies and tools only) or 'microscale'. Scale is defined in the relevant Activity Requirements or where these do not exist then per following paragraphs. Section 9.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.2) All Projects exceeding the small-scale thresholds are defined as large scale. Small scale projects are defined in accordance with CDM project standard for project activities, as below; (a) Type1: Renewable energy Projects: maximum output capacity of 15 MW(e) or 45MW(th).	parameter ICS 6 for detailed explanation). The project activity boundary is defined based on the methodology "<i>Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC)</i> (Version 4.0)" and is limited to the Republic of India (refer to section B.3 of the VPA-DD). The project activity is defined as large scale since the expected annual energy savings is 184.102GWh which exceeds 180 GWh.
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(b) Type 2: End-use energy efficiency project improvement: activities that reduce energy consumption, on the supply and/or demand side, with a maximum energy saving of 60 GWh per year (or an appropriate equivalent) in any year of the crediting period. In this context, for project activities that improve thermal energy efficiency, the maximum energy saving of 60 GWh(e) per year is equivalent to 180 GWh(th) per year saving; (c) Type 3: Other project activities: project involves technologies such Safe Water Supply, Waste management, etc. not included in Type I or Type II that result in GHG emission reductions not exceeding 60,000-ton CO₂e per year in any year of the crediting period.	
(d) Host Country Requirements Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The project activity location is the states of Gujarat, Madhya Pradesh, and Uttar Pradesh in the Republic of India. The host country has not set any mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.). A detailed assessment on the project's compliance with the applicable host Country's legal, environmental, ecological, and social regulations is summarized in table 22.
(e) Contact Details Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) As part of the Project Documentation the Project Developer shall provide (i) name	The Project Participants' contact details are provided in Appendix 2 of the VPA-DD. The project implementor is Greenway Grameen Infra Private Limited and is

and (ii) contact details of all Project Participants; AND in case of an organization (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	legally registered to operate in the host country. The company's certificate of incorporation is provided as supporting evidence.
(f) Legal Ownership Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	All end users at the time of purchase of the Greenway stove, sign a carbon rights waiver which transfers the carbon rights to the CME (Climate Impact Partners). By signing the waiver, the rights to the carbon are transferred to the CME and they remain uncontested. During the local stakeholder consultation meetings, the project implementer informed the stakeholders present the need for ownership of the carbon rights, and it was determined that the project developer shall have undisputed legal ownership to the carbon credits generated by the project activity.

Section 3.1.4 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	
(g) Other Rights Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas.	The project activity involves distribution of improved biomass cookstoves. The project activity does not involve any activity that causes alteration of any resource, or contested legal rights and other disputes, therefore the need for acquiring any specific legal right is not applicable.

h) Official Development Assistance (ODA) Declaration Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification. Section 6.1.1 and 6.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.2) Projects are ineligible for carbon crediting under Gold Standard if the ODA assistance is provided to the project under the condition that the credits generated by the Project will be transferred, either directly or indirectly, to the donor country providing ODA support. Project Developer submitting a Project located in a country named by the OECD Development Assistance Committee's ODA recipient list shall sign and submit the ODA Declaration.	The host country (India) is named in the OECD ODA recipient list¹. The project developer has signed the ODA declaration and confirms that no ODA has been provided for the project activity (<i>see provided ODA Declaration document</i>)
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¹ [DAC List of ODA Recipients for reporting 2021 flows June 2021.xlsx \(oecd.org\)](#)

i) Eligible Greenhouse Gases Section 4.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.2)</i> Only Carbon Dioxide (CO₂), Methane (CH₄) and/or Nitrous Oxide (N₂O) are eligible for Gold Standard crediting, provided Projects comply with Gold Standard Requirements and eligibility criteria.	The project activity only considers the reduction of Carbon Dioxide (CO₂), Methane (CH₄) and Nitrous Oxide (N₂O) emissions for Gold Standard crediting. Refer to section B.3 of the VPA-DD for further details.
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Corrective Action Requests (CARs) and Forward Action Requests (FARs)

FARs raised for Project Developer/CME

FAR 1. PD/CME shall make efforts to solicit input from women and marginalized groups during LSC and SFR. VVB shall assess and provide its assessment in the validation report.

The PP can confirm that women and marginalised groups were invited to provided input during the LSC and SFR consultations. As detailed in the stakeholder consultation report, invitations were sent out to organisations and groups focusing on women and other marginalised groups such as All India Women's Conference, Self Employed Women's Association and Saath Charitable Trust (see the invitation tracking sheet). Further the physical meetings were attended by both men and women, with women attendance on the first meeting 19/24 (79%). The inputs shared were addressed during the meeting as detailed on section C.3 of the stakeholder consultation report.

FAR 2. PD/CME shall undertake an upfront assessment against the Gold Standard SAFEGUARDING PRINCIPLES & REQUIREMENTS and implement their Project in accordance with the stated requirements. PD/CME shall justify and VVB shall provide its assessment in the validation report.

The PD has conducted the safeguarding principles and requirements as outlined in Appendix 1 of this VPA-DD. The PD shall make deliberate efforts to ensure the project is implemented in line with the safeguard requirements.

FAR 3. PD/CME shall take and include an expert stakeholder the expert opinion on the following SGPs:

Principle 4.1 Sites of Cultural and Historical Heritage Principle

Principle 4.2 Forced Eviction and Displacement Principle

Principle 4.3 Land Tenure and Other Rights Principle

Principle 4.4 Indigenous Peoples Principle

Principle 8.1 Impact on Natural Water Patterns/Flows Principle

Principle 8.2 - Erosion and/or Water Body Instability

Principle 9.10 - High Conservation Value Areas and Critical Habitats

Principle 9.11 - Endangered Species.

VVB shall check and provide its assessment in the validation report.

The PD conducted the safeguarding principles and requirements as required. The project activity shall entail distribution of improved cookstoves to end-users within the stated project boundary. The implementation of the project activity has not been assessed to have any negative impact on the listed safeguarding principles hence an expert stakeholder opinion is not deemed necessary.

FARs raised for validating/verifying VVB²

FAR 1. Start date of the project shall be provided as per section 4.1.39 of GS P&R. PD shall justify and VVB shall provide its assessment in the validation report.

The PD has confirmed the project start date in section C.1.1. of the VPA-DD and has provided a signed carbon waiver transfer as evidence of project start date (see evidence of start date J23VI229039).

FAR 2. VVB shall confirm that the project is not being submitted/or will be submitted/registered under any other carbon credit mechanism domestic/international/compliance/voluntary etc.).

The PD has confirmed that the project activity is not registered or seeking registration with any other carbon scheme. A declaration has been uploaded on SustainCert platform confirming the same (see GS12533 Declaration on Double Counting).

FAR 3. VVB shall confirm that the Project Area does not overlap with that of another Gold Standard or other voluntary or compliance standard program of a similar nature, is there a demonstration of no potential for double counting/misestimation of impacts.

² The PD requested for a fast-track inclusion process which was approved by SC. The project has not undergone VVB validation hence the FARs raised for the VVB have been addressed by the PD.

The PD has confirmed that the project activity has put in place measures to ensure there is no double-counting within the project activity, or other programs within the project boundary. A declaration has been uploaded on SustainCert platform confirming the same (see *GS12533 Declaration on Double Counting*).

FAR 4. VVB shall confirm that the host country, region, locality, or state does not either: Have an emission reduction cap enforced or have the possibility to trade emissions that include the scope of the proposed project.

In line with the requirements spelt out in the GS4GG Principles & Requirements (Version 1.2), Community Services Activity Requirements (Version 1.2) and GHG Emissions Reduction & Sequestration Product Requirements (Version 2.2), the PD reviewed the host country regulations. From the assessment which is detailed in table 22 of this VPA-DD, India has not enforced an emission reduction cap, and the project is not part of government's intended projects for tradeable emissions.

FAR 5. VVB to check the database management system and record keeping at CME level.

The PD has provided the project database which is uploaded on SC platform (see *GS 12533 Ex-ante ER spreadsheet, tab TSR*). The confirm required data for each project unit is maintained, and this includes the:

- Date of installation
- Geographic area of sale
- Model/type of project technology sold.
- Stove serial number
- Name, telephone number (where available), and address and/or GPS coordinates³
- LPG connection data

FAR 6. VVB to check the contracts/agreements of CME with VPA Implementers and local partners.

The CME (Climate Impact Partners) has an agreement with the VPA implementer (Greenway) and the other Project participant (SDG13 Ventures PTE. LTD). This has been uploaded on the SC platform (See *Confidential-CIP Greenway ERPA*).

³ In line with the DIGITAL PERSONAL DATA PROTECTION ACT, 2023, the project developer has complied with the provisions of this act and the database shared is a protected version with no identifiable personal information.

A.1.2. Legal ownership of products generated by the VPA and legal rights to alter use of resources required to service the project

>> All end users at the time of purchase of the Greenway stoves, sign a carbon rights waiver which transfers the carbon rights to Climate Impact Partners Limited (the CME). By signing the waiver, the rights to the carbon are transferred the CME and they remain uncontested.

A.2. Location of VPA

>> The project boundary for the proposed VPA is the Republic of India. The project activity will be implemented in three states in India which include Gujarat, Uttar Pradesh, and Madhya Pradesh. The geographical coordinates for India are Latitudes 8° 4’ and 37° 6’ north, and longitudes 68° 7' and 97° 25' east ⁴. The co-ordinates of the three states are as follows:

State	Latitude	Longitude
Gujarat	23° 00' N	72° 00' E ⁵
Uttar Pradesh	27° 40' N	80° 00' E ⁶
Madhya Pradesh	23° 30' N	80° 00' E ⁷

⁴ <https://www.india.gov.in/india-glance/profile>

⁵ [Latitude and Longitude of Gujarat, Lat Long of Gujarat \(mapsofindia.com\)](#)

⁶ [Latitude and Longitude of Uttar Pradesh, Lat Long of Uttar Pradesh \(mapsofindia.com\)](#)

⁷ [Latitude and Longitude of Madhya Pradesh, Lat Long of Madhya Pradesh \(mapsofindia.com\)](#)



Figure 1: The geographical map for the Republic of India showing location of Gujarat, Madhya Pradesh and Uttar Pradesh

Source: [Wikimedia](https://commons.wikimedia.org/wiki/File:India_location_map.svg)

A.3. Technologies and/or measures

>> The VPA involves the dissemination of Greenway Jumbo improved fuelwood cookstoves in three states in India which include Gujarat, Uttar Pradesh, and Madhya Pradesh. The improved cookstoves are manufactured by Greenway Grameen Infra private and they have been designed to improve the combustion efficiency, thereby reducing the fuelwood consumption and emissions generated while cooking. As the combustion of fuel in these stoves occurs at nearly stoichiometric conditions, the level of pollutants emitted are also reduced. The technical specifications of the Jumbo stove model disseminated under the project are given below:

Table 4: Technical specifications of the Jumbo cookstove model

Parameters	Jumbo Stove
	
Manufacturer name	Greenway Grameen Infra pvt
Size:	12.4" x 10.6" x 11.6"
Materials:	Steel and Aluminium with Bakelite Handles
Loading Capacity:	40 kg
Secondary Air Induction Mechanism:	Yes
Warranty:	2 years
Fuel Type	Ergonomic front-loading design
Operational Lifetime	72 months
Fuel type	Fuelwood
Application/service level	Domestic cooking
Energy output	2.32 KW
Efficiency	38%

Contribution to SDG benefits

The Greenway stoves are durable and efficient innovative stove technologies that are set to deliver multiple benefits to the end-users in the following ways:

- Improved health: Using traditional inefficient biomass cookstoves create negative health impacts associated with indoor air pollution along with eye & skin irritation and other ailments. Through the introduction and use of the improved project cookstoves, the households will benefit from improved health benefits due to reduced IAP related illnesses (SDG 3)

- **Environment & Climate Change:** Traditional inefficient biomass cookstoves consume more fuel, typically non-renewable firewood, or charcoal. Emissions associated with these stoves are a significant contributor to climate change. The project activity promotes use of improved cookstoves which will reduce the demand for non-renewable biomass. This will consequently reduce greenhouse gas emissions associated with use of biomass fuels for household cooking (SDG 13).
- **Gender equality:** Cooking on traditional biomass inefficient stoves consumes a lot of fuel leading to constant need for sourcing of fuel. Fuel collection requires more time and effort, especially by women and children which takes away their productive hours leading to time poverty and lost opportunity. Through reduced fuel usage as a result of using the project technology, the project activity will contribute to a reduction in time required to fetch fuel, which can be spent on other income generating activities and study for women and children (SDG 5).
- **Monetary savings:** Families spend a significant portion of their time and income on fuel collection and purchase. As fuel becomes scarce, expenditure on fuel becomes more significant due to the increase in fuel prices and this leads to more household income being spent on meeting the household energy needs. This leads to a significant reduction in household disposable income. By reducing the fuel usage through the project technology, the project activity will contribute to a reduction in the fuel cost for the households which can be spent on other household needs (SDG1).
- **Job creation:** The project activity leads to job creation in the stove production and supply chain. This is expected to benefit many people especially locals within the host country in the project areas (SDG 8).

A.4. Scale of the VPA

>> The project is a largescale project activity. As per the product activity requirements, end-use energy efficiency improvement projects are within the largescale limit if they have an annual energy saving of more than 180 GWh. As per the energy savings calculations, project activity will have an annual energy saving of 184.102GWh which is more than 180 GWh (*see GS12533-Ex-ante ER calculation tab Energy calculation*).

A.5. Funding sources of VPA

>>

There is no public funding utilized by the project. To affirm the same, the project developer has signed the ODA declaration form which has been submitted to the Gold Standard as a supporting document.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

>> The applied methodology by the VPA is “Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC), Version 4.0.”

The VPA has also applied the following latest versions of the tools and guidelines:

- Programme of Activity Requirements and Procedures, version 2.1
- Tool 30: Calculation of the fraction of non-renewable biomass, version 4.0”
- GHG Emissions Reduction & Sequestration Product Requirements, Version 2.2
- Community Services Activity Requirements, Version 1.2
- Stakeholder Consultation and Engagement Requirements, Version 2.1
- Stakeholder Consultation and Engagement Guidelines, Version 2.1
- GS4GG Principles & Requirements, Version 1.2
- Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 4.0
- Usage Survey Guidelines, version 2.0.
- SDG Impact tool, version 2.0

These are the latest versions of the methodology and tools which have been applied at the time of first submission of the project to Gold Standard.

B.2. Applicability of methodology (ies)

>> The project activity applies the GS methodology “Technologies and Practices to Displace Decentralized Thermal Energy Consumption, Version 4.0”, since the methodology is applicable to technologies and or practices that reduce or displace greenhouse emissions from the thermal energy consumption of households and non-domestic premises, it applies to this project which distributes domestic non-renewable biomass cookstoves.

The VPA involves the dissemination of improved cookstoves across India. The cooking devices being distributed are of high combustion efficiency and help in saving fuel while providing the same level of cooking service to households.

The VPA meets the applicability criteria of the methodology as follows:

Table 5: Eligibility of the project under the applicable methodology (TPDDTEC v.4.0)

Applicability condition	How Applicability condition is met by VPA
Section 2.1.1. provides that this methodology is applicable to project activities that introduce technologies and/or practices that reduce or displace greenhouse gas (GHG) emissions from the thermal energy consumption of households and/or residential, institutional, industrial, or commercial facilities. Throughout the methodology the term 'technology' should be read as the single or multiple technologies and/or practices applied in the project activity.	The project activity entails distribution of efficient biomass cookstoves. Use of these technologies will displace GHG emissions from the thermal energy consumption of households. The project is therefore eligible and emission reductions shall be calculated using method 1.
Section 2.1.2: Where there is no installation of improved devices and project claims emission reductions from improved practices only, project shall provide a detailed discussion of the chosen monitoring approach to demonstrate that quantified emission reductions result exclusively from the practices introduced by the project activity.	The project activity entails distribution of improved biomass cookstoves with higher efficiency compared to the baseline stove technologies and emissions reductions shall be claimed from improved efficiency.
Section 2.1.3: Project may involve progressive distribution of technology where implementation of the technology may occur in a gradual manner and adoption can increase over the project's crediting period.	The project activity is projected to distribute 20,000 improved cookstoves among end-users within the project boundary.
Section 2.2.1 (a) Project shall choose a technology design that has predictable performance in that it is proven to be efficient and durable under field	The rated thermal efficiency of the project stove is Jumbo Stove 38% which is above the minimum efficiency of 20% required under the meth.

conditions; for cookstoves, the rated thermal efficiency shall be at least 20%	
Section 2.2.1 (b) The technology shall have continuous useful energy output of less than 150kW per unit	The annual energy saving by the project stoves is 2.32kW which is less than 150kW.
Section 2.2.1(c) The project activity is implemented by a project developer and can include additional project participants listed in Appendix 2 of the PDD template. The individual households and institutions may be represented collectively by community organizations, etc., but do not individually act as project participants.	The project activity is being developed by Climate Impact Partners and is implemented by Greenway Grameen Infra Private Limited. Individual households are not project representatives in the project.
Section 2.2.1(d) The project developer must design incentive mechanism(s), which should be effective as fast as possible, for the elimination of inefficient baseline stoves that are replaced by the project cooking devices and describe the incentive mechanism(s) in the PDD/VPA-DD at the time of validation	The project activity has been designed with an incentive scheme that entails subsidising the cost of the stove. The Greenway Jumbo stove is priced at INR 3,499. For this project activity, the stove is being distributed to the community at a subsidised price of INR 400. Carbon financed is being utilized to support the uptake of the stoves by households. Such a facility is not available to the baseline technology users. This is an incentive mechanism which is being utilised to enable uptake of the project stoves and to discourage the use of the baseline stoves. The VPA will also monitor the extent to which the baseline technology continues to be used and where found, an appropriate adjustment carried out to account for the use of the baseline technology.
Section 2.2.1 (e) To avoid double counting or double claiming, the project developer must:	i. The VPA communicates to the end users about carbon rights during purchase of the stove and the end users sign a carbon rights waiver to transfer the carbon rights

i. clearly communicates its ownership rights and intention of claiming the emission reductions resulting from the project activity to the following parties by contract or clear written assertions in the transaction paperwork: all other project participants; project technology manufacturers; and retailers of the project technology other renewable fuel in use; and ii. inform and notify the end users that they cannot claim emission reductions from the project, and exclude from the project activity, cooking devices included in any other voluntary market or CDM project activity/PoA and strive not to displace the cooking devices of another CDM or voluntary project/PoA. See data and parameters not monitored, Avoidance of double counting or double claiming with other mitigation actions, for details on this demonstration	to the VPA developer (Climate Impact Partners Limited). ii. End-users are informed about how carbon credits are utilised to support the stove project, including the stove subsidy. They are also informed that they cannot, therefore, claim emission reductions from the project. The project stoves are identified by their serial numbers through which the stoves are uniquely traceable to the end users based on the sales database records. The project implementer ensures that the sales database is well maintained and updated and only the serialised stoves in the sales database are included for emission reduction calculations. This way, double counting is avoided for the project stoves.
Section 2.2.1 (f) Project activities making use of solid fossil fuel in the project scenario or other improved fossil fuel cookstoves meeting certain conditions described in the footnote to Table 1 (e.g. switch from three-stone fire biomass stoves to LPG stoves) may only claim emission reductions for energy efficiency improvement aspect and shall assume the same baseline and project fuel for emission reduction calculations	Not applicable to the project. The project only uses non-renewable biomass.
Section 2.2.1 (g) Project activities making use of a new solid biomass feedstock in the project situation (e.g. switch to green charcoal or renewable biomass briquettes)	Not applicable to the project. The project only uses non-renewable biomass.

must comply with relevant specific requirements for biomass related project activities, as defined in the latest version of the Community Services Activity Requirements. The specific requirements apply to both plantations established for the project activity and/or existing plantations that will supply biomass feedstock	
Section 2.2.1 (h) Adequate evidence is supplied to demonstrate that indoor air pollution (IAP) levels are not worsened compared to the baseline, and greenhouse gases emitted by the project fuel/stove combination are estimated with adequate precision. Furthermore, for projects where cooking will move from outdoor to indoor or where the project technology reduces ventilation (for example, changing from a stove with chimney to improved stove with no chimney), indoor air pollution (IAP) levels shall not worsen in the project compared to the baseline, including PM 2.5 and carbon monoxide (CO) emissions. This may be demonstrated before project Design Certification or during project operation using the certification resulting from of a manufacturer's test, report of field testing of the technology's PM 2.5 and carbon monoxide (CO) emissions, report of lab testing of the technology, or results of modelling of the technology's operation under field conditions. If none of these are available, reference from published literature or report by independent agencies may be used as	The Greenway stoves are portable, it is possible for the cooking to be moved from outdoor to indoor. The project stove has been tested for PM2.5 as a way of determining the indoor air pollution levels. This will be compared with the PM 2.5 levels of the baseline stoves to be determined from published literature. Furthermore, the project will not reduce ventilation in the kitchens as they are improved stoves that will contribute to improved indoor air conditions among beneficiary households.

evidence, provided it is not more than 5 years old.	
Section 2.3.1 The project shall not undermine or conflict with any national, sub-national or local regulations or guidance for thermal energy supply or fuel supply or use. The project shall document the national, regional and local regulatory framework for provision of thermal energy services of the type of the project provided in the project boundary (parameter ICS 7).	There are no regulations in India that require households to adopt efficient fuelwood stoves. However, The Ministry of New and Renewable Energy approves models of portable improved biomass cookstoves, which users can purchase (See Section B-4 of the VPA-DD). One of the approved stoves models is the Greenway stove (see evidence; List of Approved Cookstoves by the Ministry).
Section 2.3.2 If the expected technical life of project technology (parameter ICS 3) is shorter than the crediting period, the project developer shall describe measures to ensure that end users are provided replacement technology of comparable quality at the end of the technical life, by either replacing with comparable or better technology, or retrofitting essential parts with performance guarantee. If neither of the prior conditions can be demonstrated, no emission reductions can be claimed for the technology after its technical life has ended.	The technical life of the project stove is 6 years which is more than the GS crediting period. Therefore, there will be no replacement of the technology before the end of the crediting period.

B.3. VPA boundary

>>

The project boundary of the proposed project encompasses the following:

- a. The physical and geographical sites within the states where the Greenway Jumbo stoves will be utilized within households.
- b. The target area of the project is Gujarat, Uttar Pradesh, and Madhya Pradesh in India.

The assessment of sources and gases included in the project boundary is given below.

Table 6: Assessment of GHG in the project boundary

Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Delivery of thermal energy (fuelwood on inefficient cookstoves)	CO ₂	Yes	Important source of emissions
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Important source of emissions
Project scenario	Delivery of thermal energy (fuelwood using efficient cookstoves for cooking)	CO ₂	Yes	Important source of emissions
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Important source of emissions

B.4. Establishment and description of baseline scenario

>> According to the methodology applied by the project activity, TPDDTEC, version 4.0, the baseline scenario is defined by the existing baseline technology/practice use and fuel consumption patterns for the type of service provided by the project technology in the population targeted for adopting the new project technology. Hence, this “target population” is a representative baseline for the project activity.

The selection and description of the baseline scenario has been informed by the findings of the Baseline Scenario Survey conducted for the project activity. The baseline scenario survey was conducted with 187 households who were randomly selected from the project target areas, and who were not using the project technology.

The surveys were conducted between 20/09/2023 to 29/09/2023 by the project implementor (Greenway).

The surveys were done in the three states of Gujarat, Uttar Pradesh, and Madhya Pradesh in India where the stoves are going to be distributed.

Summary of baseline surveys by State

State	No of surveys
Gujarat	58
Uttar Pradesh	65
Madhya Pradesh	64
Total	187

Out of the 187 surveys conducted, all the households (100%) were using firewood for cooking as the primary fuel, using traditional 3 stone firewood stoves.

From the baseline survey conducted, it was determined that the targeted communities use non-renewable biomass in traditional inefficient stoves. The default efficiency of such stoves as per applied methodology is 10%. The baseline scenario has therefore been determined as, the consumption of non-renewable firewood using traditional inefficient stoves to meet thermal energy requirements for household cooking.

However, as part of the installation survey, the project implementer assessed usage of the host government LPG program. This data is recorded on the sales database and is analysed on the ER sheet tab LPG assessment. It was observed that although some end-users are connected to the LPG network, they hardly use it for cooking. To determine the relevance of LPG consumption in the emission reductions, end-users were requested to confirm how often they refill their LPG per year. End-users who refill their LPG three time and above are considered to be stacking and their usage is discounted in the emission reduction calculations. The LPG consumption will be monitored as part of the usage survey and relevant discount factor shall be applied.

The state wise breakdown of LPG connection and applied ex-ante discount factor is as follows:

Table 7: Analysis of LPG connection among project end-users

Gujarat		
LPG connection	Number of households	% discount
Yes	801	33.1%
no	1618	66.9%
Total	2419	100%
Madhya Pradesh		
LPG connection	Number of households	% discount
Yes	7	1.9%
no	365	98%
Total	372	100%
Uttar Pradesh		
LPG connection	Number of households	% discount
Yes	1	1.3%
No	79	98.8%
Total	80	100%

a) Baseline scenario survey

The baseline scenario survey provides critical information on target population characteristics, baseline technology use, fuel consumption, leakage, and sustainable development indicators.

The Project developer undertook a baseline scenario survey to collect the required baseline data from the target population, baseline technology use and fuel consumption, and the sustainable development indicators. The survey was conducted on the period between 20/09/2023 to 29/09/2023 by the project implementor (Greenway) and interviewed 187 households.

b) Representativeness

The baseline survey requires in-person interviews with a robust sample of end-users without project technologies that are representative of end-users targeted in the project activity.

The project implementer conducted in-person interviews with a sample representative of the end-users targeted by the project activity. The baseline surveys were conducted within the applicable geographical areas covered under the project boundary.

The households surveyed were randomly selected and it was established that they were not using the project technology by the time of the survey. The surveys were done in the three states where the stoves will be distributed.

c) Sample Sizing: The methodology requires that baseline survey should be carried out for each baseline scenario using representative and random sampling, following these guidelines for minimum sample size:

Group size	Minimum sample size
<300	30 or population size, whichever is smaller
300 to 1000	10% of group size
> 1000	100

The targeted population for the group activity is 20,000 households. Since the group size is more than 1000, the minimum sample required shall be 100 households without using the project technology. In person interviews were held with the 187 household who were not using the project technology, to collect the required baseline data. The extra households were to cater for any sample size attritions.

The minimum sample size per state was calculated as a proportion of the number of the projected stoves to be distributed in each state. A summary of the selected samples and the actual surveys conducted per state is provided below.

Table 8: Baseline survey sampling

State	Stove population	% stove distribution	Baseline survey sample	Actual surveyed
Gujarat	5,717	28.59%	43	58
Uttar Pradesh	7,993	39.97%	60	65
Madhya Pradesh	6,290	31.45%	47	64
Total	20,000	100%	150	187

Source: GS12533 Baseline and KPT sampling

d) Data collected:

Information on the following was gathered during the surveys:

1. Users follow up	i. Address or location ii. Mobile telephone number and/or landline telephone number (when possible)
2. End user characteristics	iii. Number of people served by baseline technology. iv. Typical baseline technology usage patterns and tasks (commercial, institutional, domestic, etc.)
3. Baseline technology and fuels	v. Types of baseline technologies vi. Types of fuels used and estimated quantities. vii. Sources of fuels; purchased, hand-collected, or etc., and prices paid, or effort made, e.g. walking distances, persons collecting, opportunity cost viii. Fuel collection responsibilities and cooking behaviours

e) Baseline scenario survey findings

1. End-user characteristics

The baseline survey was conducted with 187 households, who included 21.9% men and 78.1% women. From the survey it was determined that women are the majority (96.3%) primary cook, with men only being 3.7%. The average household size was calculated at 6.0.

Table 9: Gender analysis of survey respondents and primary cook

Gender analysis				
	Interviewees		Primary cook	
	outcome	%	outcome	%
Male	41	21.9%	7	3.7%
Female	146	78.1%	180	96.3%
Total	187	100.0%	187	100.0%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

2. Baseline fuels, stove technologies, and cooking practices

As summarised in table below the main source of the firewood used in the baseline was identified as collecting from nearby forest/wastelands (71.1%), while 28.9% of the households indicated they buy from local vendors.

Table 10: Fuel acquisition by households in the baseline

Source of baseline fuel		
parameter	outcome	%
Purchase from local vendors	54	28.9%
Collecting from forest or woodlands	133	71.1%
Total	187	100.0%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

On fuel acquisition responsibilities, 58.3% reported it was the responsibility of women to get the fuel compared to men at 41.7%.

Table 11: Fuel collection responsibilities in the baseline

Gender of person obtaining fuel		
parameter	outcome	%
Male	78	41.7%
female	109	58.3%
Total	187	100.0%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

In terms of time spent in getting the baseline fuel, majority of the households (62.6%) reported spending more than 1 hours per day in fuel collection.

Table 12: Time spent in fuel collection in the baseline

Time spent per day in fuel collection		
parameter	outcome	%
<30 minutes	6	3.2%
30 minutes-1 hour	64	34.2%
1-2 hours	71	38.0%
>2 hours	46	24.6%
Total	187	100.0%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

The analysis of the primary baseline stove technologies is presented in table below. The frequency of use of the stoves was rated as daily by 58.8% of the respondents, while 24.1% reported using the stoves for 3-6 days in a week. Only 17.1% used it less than 2 days in a week.

Table 13: Frequency of use of the primary baseline stoves

Frequency of baseline stove usage per week		
parameter	outcome	%
1 day	16	8.6%
2 days	16	8.6%
3 days	20	10.7%
4 days	11	5.9%
5 days	4	2.1%
6 days	10	5.3%
7 days	110	58.8%
Total	187	100.0%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

The time spent on cooking using the baseline stove was also assessed where majority (41.2%) spent between 45-60 minutes. 10.2% spent more than 2 hours, while 8% spent less than 45 minutes. The average number of meals cooked per day was calculated at 2.2.

Table 14: Time spent cooking

Average time taken to cook a single meal		
parameter	outcome	%
<45 minutes	15	8.0%
45-60 minutes	77	41.2%
1-2hours	76	40.6%
>2 hours	19	10.2%
Total	187	100.0%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

Majority of the households (60.4%) cook within the main house, with only 1.1% having a building separate from the main house.

Table 15: Cooking areas with the baseline stoves

Cooking area with baseline stove		
parameter	outcome	%
Inside main house	113	60.4%
Outside near to the house or in a courtyard	39	20.9%
On a veranda or porch	33	17.6%
Inside a building separate from the main house	2	1.1%
Total	187	100.0%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

On seasonal variation of the baseline fuel consumption, 33.2% spent less during dry seasons while 58.8% spent more during the rainy season. During holidays and weekends 69.5% and 54.5% of the respondents reported using more fuels respectively.

Table 16: Assessment of baseline fuel consumption based on seasonal variation.

	Rainy seasons		Dry seasons		Weekends seasons		Holiday/Festivities seasons	
parameter	outcome	%	outcome	%	outcome	%	outcome	%
More use	110	58.8%	100	53.5%	102	54.5%	130	69.5%
less use	58	31.0%	63	33.7%	53	28.3%	43	23.0%
No change	19	10.2%	24	12.8%	32	17.1%	14	7.5%
Total	187	100%	187	100%	187	100%	187	100%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

3. Baseline Sustainable development assessment

The survey respondents were asked to rate the current level of indoor smoke within their households. A majority (71.7%) expressed the smoke level range was high while only (1.6%) expressed the level as low.

Table 17: Analysis of smoke levels in households

Current level of smoke		
parameter	outcome	%
High	134	71.7%
Moderate	50	26.7%
Low	3	1.6%
total	187	100%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

The respondents were also asked about the incidence of smoke related illnesses using the baseline technology. Only 23.5% of the respondents had not suffered an illness, with majority (76.5%) reporting suffering smoke related illnesses such as itchy eyes or eye problems, coughing and respiratory illnesses.

Table 18: Incidence of smoke related illnesses

Incidence of smoke related illnesses		
parameter	outcome	%
Itchy eyes or eye problems	61	32.6%
Coughing	76	40.6%
Respiratory illness	6	3.2%
none	44	23.5%
total	187	100%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

The level of satisfaction in using the baseline stoves was assessed among the households surveyed. The majority at 46.5% expressed dissatisfaction in using the baseline stoves while 42.8% were partly satisfied. Only 10.6% expressed some satisfaction with their baseline stoves.

Table 19: Assessment of the level of satisfaction in using the baseline stoves.

Level of satisfaction using baseline stove		
parameter	outcome	%
More than satisfied	1	0.5%
Very Satisfied	4	2.1%
Satisfied	15	8.0%
Partly Satisfied	80	42.8%
Not at all Satisfied	87	46.5%
total	187	100%

Source: Baseline survey data spreadsheet tab consolidated baseline survey

The project activity is a non-industrial activity and only a single technology is installed in a household. In line with the applicable methodology, the baseline is fixed until the end of the crediting period and will not require continuous monitoring or updating. If a project cookstove is replaced with a cookstove of similar efficiency prior to the end of the crediting period, the original baseline shall be applicable till the end of the replaced cookstoves’ useful lifetime or the registered crediting period, whichever occurs earlier.

The proposed VPA, will disseminate Greenway Jumbo cookstoves with a thermal efficiency of 38%, which will replace the baseline stoves currently used for domestic cooking by the target communities. This action will lead to a reduction in the use of non-renewable biomass, and this will be the new project scenario. In the project scenario, highly efficient stoves are used in households, which consume less non-renewable biomass compared to the baseline scenario.

4. Baseline non-renewable biomass (NRB) assessment

The baseline scenario identified by the project activity assumes use of non-renewable biomass (firewood) as the main baseline fuel among the targeted population. The fractional non-renewability of biomass has been established using the latest CDM tool, “Tool 30: Calculation of the fraction of non-renewable biomass, version 4.0”.

The project developer has conducted state wise calculation of the fNRB value using the most recent available forestry data for the 3 states. The calculations and data sources are detailed in the ex-ante ER sheets. The calculated values are as follows:

Table 20: Calculated fNRB values per state

State	fNRB
Gujarat	97.08%
Madhya Pradesh	92.09%
Uttar Pradesh	99.03%

Source : GS12533 Ex-ante ER sheet tabs Data parameters

The value of fNRB is fixed ex-ante for the entire crediting period.

5. Baseline and Project Fuel Consumption

The quantity of fuel consumed in the baseline and project scenario was determined based on the average annual consumption of firewood per household, established through a KPT carried out by the project implementer.

The project carried out a paired KPT test to determine fuel consumption by the end-users first when using the baseline cookstove, and then while using the project cookstove.

The Kitchen Performance Test Procedure stipulated under Annex 2 of the applicable methodology was applied.

Summary of the procedure followed:

The fuel measurement involved a paired test where a household used baseline stove for 3 days and after 3 days, that household switched from using the traditional stove to the Greenway Jumbo stove which they used for the 3 days.

Procedure followed:

1. Households to participate in the test were selected through a random sample from the stoves database.
2. The purpose of the KPT was well explained to the household members
3. Measurement of firewood usage for each day schedule was agreed on and the households were encouraged not to change their cooking behaviour/patterns throughout the test period and not to give away any firewood to relatives/friends as this would lead to disqualification from the survey.
4. Inventory area for fuel consumption measurement was well defined. Weighed firewood for the KPT were stored in a safe area for easy monitoring.
5. On day 1, the household background information was captured in the questionnaire form.
6. Testing periods of 3 consecutive days were all agreed on.
7. Households were visited daily, without being intrusive; and the amount of firewood used was weighed.

Sampling criteria

The number of end-users to be sampled for the KPT was calculated by the CME following the Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 4.0. The KPT sample size was determined as follows:

Sample Size Determination for the Survey and Estimation of Utilisation		
Sample Size Calculation		
$n \geq t^2 \cdot N \cdot (SD^2/p^2) / (((N-1) \cdot 0.1^2) + (t^2 \cdot (SD^2/p^2)))$		
<i>n</i>	43	Minimum size of the sample
<i>N</i>	20,000	Size of the population
<i>t</i>	1.96	Confidence interval (taken as 1.645 and 1.96 for 90% and 95% confidence intervals, respectively)
<i>p</i>	0.9	Population proportion, set at 0.90 (the proportion of stoves still in use after 3 years assuming an annual drop off rate of 5%)
<i>l</i>	0.1	Represents the 10% relative precision
<i>SD</i> ²	0.09	Overall variance

$$SD^2 = \frac{(g_a \times p_a(1-p_a)) + (g_b \times p_b(1-p_b)) + (g_c \times p_c(1-p_c)) + \dots + (g_k \times p_k(1-p_k))}{N} \quad (44)$$

Stove Sizes	Stove Numbers (g _a)	Estimated Stoves in Operation (p _a)	SD ²
Gujarat	5717	0.9	0.03
Uttar Pradesh	7,993	0.9	0.04
Madhya Pradesh	6,290	0.9	0.03
Total Sales	20,000	0.9	0.09

The samples were proportioned state wise based on the number of stoves distributed in each state. The sample distribution was as follows:

State	No. of stoves	% distribution	Proportional samples	Oversampled by 20%
Gujarat	5717	28.6%	12	15
Uttar Pradesh	7993	40.0%	17	21
Madhya Pradesh	6290	31.5%	14	16
Total	20000	100%	43	52

A minimum sample of 43 was determined to be sufficient to meet the required sample size. The CME oversampled by 20% and the sample derived was 52. The extra samples were to cater for any sample attritions during the survey. The samples were randomly selected from the 187 end-users who took part in the baseline survey. Since the KPT was paired, the population that took part in the baseline and project KPT was the same. Once the required sample size was determined, an online random sample selection application, Research randomizer⁸ was used to randomly select the end-users from the baseline survey respondents.

Screenshots from Gujarat sampling

⁸ [Research Randomizer](#)

← ↻ 🔍 https://www.randomizer.org/#randomize

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Do you wish each number in a set to remain unique? Yes ▾

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RESULTS

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1 Set of 15 Unique Numbers
Range: From 1 to 67

Set #1
12, 65, 27, 26, 53, 23, 57, 39, 9, 63, 64, 66, 47, 52, 14

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Source: GS12533 Baseline and KPT sampling tab Gujarat

Screenshots from Madhya Pradesh sampling

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How many numbers per set?

Help
Specify how many numbers you want Research Randomizer to generate in each set. For example, a request for 5 numbers might yield the following set of random numbers: 2, 17, 23, 42, 50.

Number range (e.g., 1-50)

Help
Specify the lowest and highest value of the numbers you want to generate. For example, a range of 1 up to 50 would only generate random numbers between 1 and 50 (e.g., 2, 17, 23, 42, 50). Enter the lowest number you want in the "From" field and the highest number you want in the "To" field.

Source: GS12533 Baseline and KPT sampling tab MP

Screenshots for Uttar Pradesh sampling

← ↻ 🔍 https://www.randomizer.org/#randomize

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RESEARCH RANDOMIZER

RANDOMIZE TUTORIAL LINKS ABOUT

How many sets of numbers do you want to generate?

Help

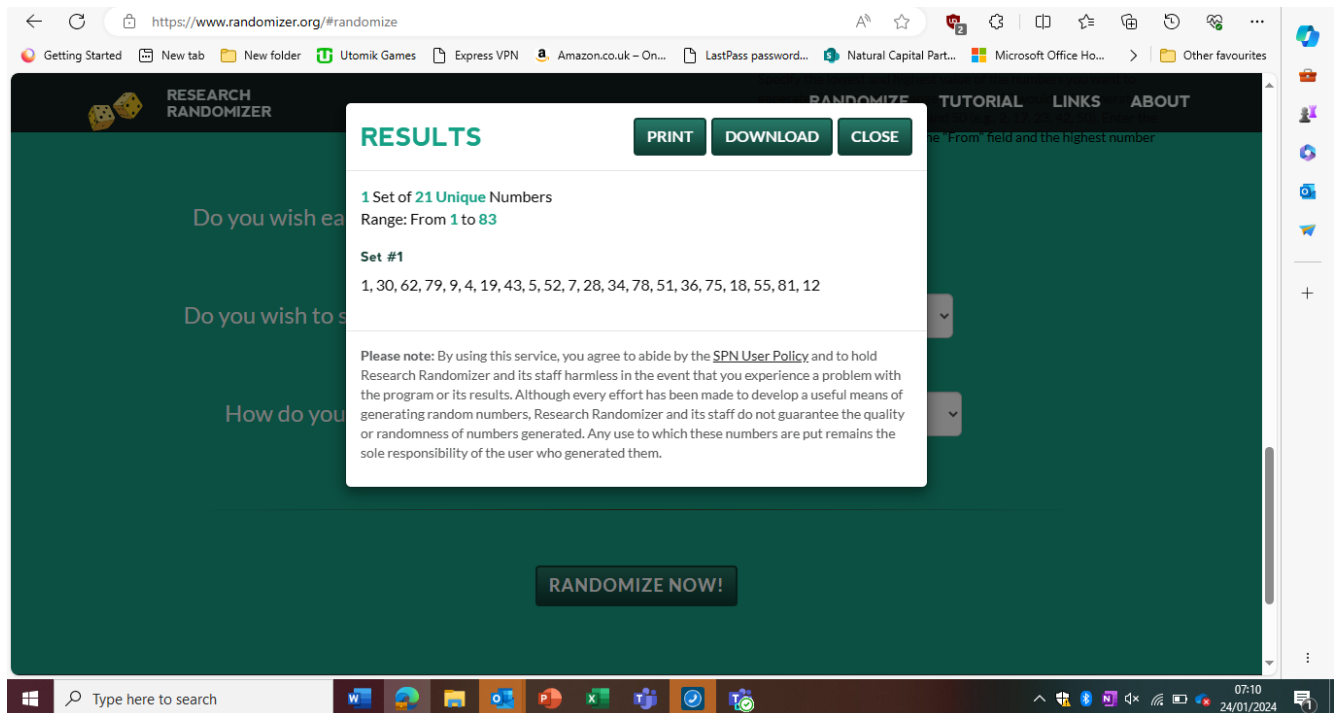
How many numbers per set?

Help

Number range (e.g., 1-50)

Help
Specify the lowest and highest value of the numbers you want to generate. For example, a range of 1 up to 50 would only generate random numbers between 1 and 50 (e.g., 2, 17, 23, 42, 50). Enter the lowest number you want in the "From" field and the highest number you want in the "To" field.

Do you wish each number in a set to remain unique?



Source: GS12533 Baseline and KPT sampling tab UP

The CME conducted a cross-check to confirm the required 90/10 precision was met with the KPTs conducted using the GS KPT analyser tool. The analysis from the KPT analyser tool is as follows:

Baseline KPT single fuel sample test

Step -1 Outlier removal

Outlier identification	
Mean	7.924
Standard Deviation	2.47
COV	0.31
Upper Quartile	8.72
Lower Quartile	6.21
Interquartile Range	2.51
Outlier Threshold (Upper)	12.49
Outlier Threshold (Lower)	2.45

Step -2	90/10 assessment	
<i>Sample Mean</i>	7.425	7.425
<i>Sample Size (valid count)</i>		48
<i>Standard Deviation</i>		1.80
<i>Standard error</i>		0.26
<i>What is the precision attained?</i>		4%
<i>Does the result satisfy the 90/10 rule?</i>		YES
<i>Fuel consumption</i>	<i>Use mean value</i>	
<i>Mean value</i>	<i>kg/per household/day</i>	<i>7.425</i>
<i>Lower bound</i>	<i>kg/per household/day</i>	

Step -3 Result

Mean value	7.425	kg/hh/day
------------	-------	-----------

Source : GS12533 KPT analyser tab KPT baseline

Project stove (Greenway Jumbo) KPT single sample test

Step-1: Outlier Identification	
Mean	3.56
Standard Deviation	1.35
COV	0.38
Upper Quartile	4.25
Lower Quartile	2.59
Interquartile Range	1.66

Step -2	90/10 assessment	
<i>Sample Mean</i>		<i>3.42</i>
<i>Sample Size (valid count)</i>		<i>50</i>
<i>Standard Deviation</i>		<i>1.18</i>
<i>Standard error</i>		<i>0.17</i>
<i>What is the precision attained?</i>		<i>6%</i>
<i>Does the result satisfy the 90/10 rule?</i>		<i>YES</i>
<i>Fuel consumption</i>	<i>Use mean value</i>	
<i>Mean value</i>	<i>kg/per household/day</i>	<i>3.4239</i>
<i>Lower bound</i>	<i>kg/per household/day</i>	

Step -3 Result

Mean value	3.42	kg/hh/day
------------	------	-----------

Source: GS12533 KPT analyser tab KPT baseline

6. Relevant applicable legislation and how effectively these are enforced (GS4GG Principle 1).

In India, there is no requirement that households must purchase efficient cookstove or any project which targets to implement a domestic household project must install efficient cookstoves. This, therefore, is a voluntary action by the project developer to initiate this project. The project is not being implemented as a result of any existing regulatory requirements. In addition, the Ministry of New and Renewable Energy provides a list of all approved improved cookstoves which can be distributed in the country and the Greenway stoves are among those stoves which have been listed as approved by the ministry.

B.5. Demonstration of additionality

>>

Section 3.3.1 of the applied methodology, TPDDTEC Version 4.0 provides that “The project developer must show that the project could not or would not take place without the presence of carbon finance”.

Further, the project developer shall demonstrate additionality prior to project registration by conforming to additionality requirements of one of the options below,

- a. Applicable GS4GG Activity Requirements.
- b. CDM Tool 01 - Tool for the Demonstration and Assessment of Additionality.
- c. CDM Tool 19- Demonstration of additionality of microscale project activities; (not applicable to Gold Standard microscale projects)
- d. CDM Tool 21 – Demonstration of additionality of small-scale project activities; (applicable to small-scale projects only)
- e. An approved Gold Standard VER additionality tool

Pursuant to section 4.1.9 of Community activity service requirements, “Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification”.

(a) Positive list

(b) Projects located in LDC, SIDS, LLDC4

(c) Microscale projects

The Community Services Activity Requirements V1.2, Annex B on positive list, section 1.1.5 lists *“Project activities that involve technologies and/or practices providing thermal energy to the user that have less than 20% adoption rate among the target users”* as additional. The additionality of the project has been established using this provision.

The project activity entails distribution of improved firewood cookstoves to deliver thermal energy to households for domestic cooking. The primary targets are end-users using inefficient biomass stoves especially those in rural areas.

According to the *State of Clean Cooking Energy Access in India* report published by CEEW in September 2021⁹, half of Indian households continue to use firewood, and nearly a fourth of the households reported using dung cakes and other solid fuels for cooking despite improvements in LPG use. Most solid fuel users are in rural India, where 67% of the households use firewood for cooking. This practice is a concern due to

⁹ [ires-report-on-state-of-clean-cooking-energy-access-in-india.pdf](https://www.ceew.in/res-report-on-state-of-clean-cooking-energy-access-in-india.pdf) (ceew.in)

households continued high exposure to Household Air Pollution (HAP). The *Roadmap for Access to Clean Cooking Energy in India* report by the Council on Energy, Environment and Water (CEEW) published in 2019¹⁰ reports that, less than 1% of rural households surveyed reported using improved biomass cookstoves (ICS). Only 14% of households surveyed in the report were aware of their existence, indicating low awareness among non-users (refer to section 5.3 on page 38).

The Government of India have established policies in the past to increase the penetration rate of improved cookstoves. These programmes include the National Programme on Improved Chulhas (NPIC), which sought to introduce 35 million chulhas between 1986 and 2002 (MNES, 2004) and the National Program on Biogas Development (1982). However, studies show that the current use of these technologies is low. A 2016 survey by Practical Action in six states (including Uttar Pradesh and Madhya Pradesh) indicates that less than 1 per cent¹¹ of the rural population was using these technologies.

The other program introduced by the Government in 2014 was the Unnat Chulha Abhiyan (UCA), with the aim of deploying 2.75 million ICS by March 2017 (MNRE, 2014b). However, the report by Council on Energy, Environment and Water (CEEW) in 2019 indicate that the scheme had met only 1.3 per cent of its target by March 2017, with much of the budget having lapsed unutilised.

In 2016, the Government of India also launched the *Pradhan Mantri Ujjwala Yojana* (PMUY) or *Ujjwala* scheme which seeks to distribute more than 80 million LPG connections. However, study shows that having an LPG connection does not ensure its sustained use by households. Households are still stacking LPG with solid fuels due to limited affordability, lack of timely availability of LPG cylinders, availability of free biomass, and cultural preferences propel households to stack LPG with solid fuels. Most solid fuel users are in rural India, where despite 80 per cent of households having an LPG connection, 67 per cent use firewood for at least some of their cooking needs.

Based on the above discussions, it is evident that significantly less than 20% of Indian households that use biomass for cooking use non-improved biomass cookstoves. The additionally criteria set out in section 1.1.5 is therefore met and the project activity is deemed additional.

¹⁰ [ResearchGate](#)

¹¹ [ceew-research-energy-poverty-measuring-energy-access-india.pdf](#)

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).

As per GS4GG Community services activity requirements, Version 1.2, Para 4.1.9, Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of design certification:

- (a) Positive list (Annex B of this document)
- (b) Projects located in LDC, SIDS, LLDC
- (c) Microscale projects

Describe how the proposed VPA meets the criteria for deemed additionality.

The project meets criterion 1. Positive list, Annex B of the Community Services Activity Requirements. Section 1.1.5 of Annex B lists "Project activities that involve technologies and/or practices providing thermal energy to the user that have less than 20% adoption rate among the target users" as additional. As discussed above, the adoption rate of improved biomass stoves in India states is less than 20% and hence the project is additional.

B.5.1. Prior Consideration

>> According to the Gold Standard's Principles and Requirements (version 1.2 para 4.1.49), retroactive projects shall submit the required documents for preliminary review (time of first submission) within one year of the project start date. Retroactive Project submitted at a date later than one year from the project start date will not be eligible for Gold Standard certification.

The first stakeholder meeting was held on 10/11/2023 and the second was held on 20/11/2023. The stove distribution under the project activity started on 16/10/2023 which was after the first consultation. The project cycle has therefore been identified as retroactive and the project is hence required to demonstrate prior consideration. The first submission to the GS was made on 09/01/2024 when the LSC documentation was uploaded for preliminary review, and this is within 1 year from the start of the project.

The project is hence eligible for Gold Standard certification and the key milestones are as summarized below.

Table 21: Key project milestones

Timeline	Milestone
10/11/2023	The first LSC meeting
16/10/2023	Start date of the project (first stove distribution).
20/11/2023	Second LSC meeting
09/01/2024	The first submission date to GS
24/01/2024 - 24/03/2024	Stakeholder feedback round (SFR)
27/02/2024	Project listing

Further, in line with para. 7.1.3 of GHG Emissions Reduction & Sequestration Product Requirements v.2.2. *"If the stakeholder consultation for the Project was conducted after the start date of the Project, the Project Developer shall demonstrate that the revenues from carbon credits were seriously considered in the decision to implement the Project."*

The project developer signed a funding agreement for the implementation of the project activity with the project implementer on 30th August 2023, which is before the start date of the project (16/10/2023). The funding agreement provides confirmation that carbon revenue had been considered from the beginning before implementation of the project. The relevant supporting document has been uploaded on SustainCERT platform (ref. CONFIDENTIAL CC-Greenway_Funding_and_ERPA_CIP_30082023).

B.5.2. Ongoing Financial Need

>> Not applicable. As per para 4.1.52 of Principles and Requirements¹², Ongoing Financial Need shall be demonstrated at Design Certification Renewal.

¹² [101 V1.2 PAR Principles-Requirements \(goldstandard.org\)](#)

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
13 Climate Action	13.2 Integrate climate change measures into national policies, strategies, and planning.	Amount of Verified Emission Reductions (VERs)
1. No Poverty	1.1 By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 a day	Estimated monetary savings computed from fuel saved
5. Gender Equality	5.4: Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	Average time saved by women in fuel collection.
7. Access to clean energy	7.1: By 2030, ensure universal access to affordable, reliable and modern energy services	Number of people/households with access to basic service provided by the project device
8. Decent Work and economic growth	8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Total number of jobs split by gender

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

>> **Methodological choices/approaches related to SDG 13**

Since the baseline fuel and project fuel are the same, emission reductions are exclusively from improved efficiency. The project, therefore, shall use Method 1 in calculating the emission reduction. Therefore, the GHG emissions reduction achieved by the project activity in year y is calculated using equation 1.

In line with the applied methodology TPDDTEC, version 4.0, emission reductions can be estimated by using the following formula.

$$ER_y = \sum (b_{,p} Nb_{,p,y} \times Up_{,y} \times SFSp_{,b,y} \times NCVb_{,fuel} \times (fNRB_{,b,y} \times EFb_{,f,CO2} + EFb_{,f,nonCO2})) - \sum LEp_{,y} \dots Eq1$$

Where:

Parameter	Description	Value applied	Unit	Data source
$\sum_{b,p}$	Sum over all relevant (baseline b/project p) couples	0.6689(Gujarat) 0.9812 (Madhya Pradesh) 0.9875 (Uttar Pradesh)	fraction	Calculated to discount LPG users (values achieved from installation survey)
$Nb_{,p,y}$	Number of project technology-days included in the project database for baseline b/project p pair in year y (days)	1	Number	Internal assumption for ex-ante purposes
$Up_{,y}$	Cumulative Usage rate for technologies in project scenario p in year y (fraction)	0.90	Fraction	Internal assumption for ex-ante purposes
$SFSp_{,b,y}$	Specific fuel savings for an individual project technology of baseline b/project p pair in year y (mass or volume units/technology*day)	0.0040	tons /hh/ year	Calculated from KPT results
$f_{NRB,b,y}$	Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass (drop this term from the equation when using a fossil fuel baseline scenario)	0.9708 (Gujarat) 0.9209 (Madhya Pradesh) 0.9903 (Uttar Pradesh)	Fraction	Calculated

NCV _{b,fuel}	Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.015 TJ/ton)	0.0156	TJ/ton	Default value as per applied methodology
EF _{b,fuel,CO2}	CO ₂ emission factor of the fuel that is substituted or reduced. 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel	112	tCO ₂ /TJ	Default value as per applied methodology
EF _{b,fuel,nonCO2}	Non-CO ₂ emission factor arising from use of fuel f when the baseline fuel f is biomass or charcoal (tCO ₂ e/TJ). This parameter is omitted when f is a fossil fuel.	9.46	tCO ₂ /TJ	Default value as per applied methodology (AR5 GWP)
LE _{p,y}	Leakage for project scenario p in year y	0.0811 (Gujarat) 0.1133(Madhya Pradesh) 0.1219(Uttar Pradesh)	tCO ₂ e/yr	Calculated applying the default value of 0.95
ER _y	Emission reduction for total project activity in year y	1.54 (Gujarat) 2.152 (Madhya Pradesh) 2.316 (Uttar Pradesh)	tCO ₂ e/yr	Calculated

Source: GS12533 Ex-ante ER spreadsheet tab data parameters

Leakage

The leakage emissions, $LE_{p,y}$, for the project is determined using Option 1, whereby a default adjustment factor of 0.95 to the emission reductions to approximate leakage emissions is used in the ER estimation.

Methodological choices/approaches related to SDG 1

The VPA includes the distribution of fuel-efficient cookstoves to replace the inefficient baseline cooking methods and thus result in a reduction in firewood consumption. The provision of the project technology coupled with the reduced fuel demand for the daily cooking activities increases access to affordable energy and improved cooking technologies (i.e. basic services) for the stove users. This also enables households to spend less on fuel purchases, therefore, making savings.

The contribution of the project to this target of SDG 1 will be measured by calculating the estimated household savings based on the prevailing fuel prices. This will be calculated as follows:

$$\text{Net Benefit (SDG 1)} = \text{Financial savings}_{\text{baseline}} - \text{Financial savings}_{\text{project}}$$

Monetary savings	Outcome
Average daily expenditure on fuel in the baseline (R)	37.082
Expenditure with project stove	9.76
Savings (Rupee)	27.32

Source: GS12533 Ex-ante ER spreadsheet tab specific savings

Methodological choices/approaches related to SDG 5

The use of a Greenway stove in a household helps in reducing the time spent on firewood collection which is mostly done by women. Fuel collection takes more time and effort, which takes away productive hours of women and young girls leading to time poverty and lost opportunity.

The average time saved by women during fuel collection will be determined as follows:

Average fuel collection time = the average number of times women in a household used to fetch firewood in the baseline scenario – the average number of times women in a household fetch firewood in the project scenario.

This is calculated as follows:

Time saving	
Average daily time spent in baseline fuel search (hrs)	1.46
Average daily time spent in project fuel search (hrs)	0.38
Savings (hrs)	1.07

Source: GS12533 Ex-ante ER spreadsheet tab specific savings

Methodological choices/approaches related to SDG 7

Parameter: Number of people/households with access to basic service provided by the project device

This parameter will be monitored based on the actual number of operational units sold as indicated in the sales database. The project implementor shall maintain an updated sale database for all the units sold by the project.

The project is expected to distribute 20,000 improved fuelwood cookstoves among household end-users.

Methodological choices/approaches related to SDG 8

The VPA is expected to create job opportunities through the stove production and distribution process. This parameter will be monitored through project reports on the number of people employed. The reporting will also include data on the number of males and number of females involved in the project.

In the baseline there were no people employed by the project. In the project scenario, the project targets to employ 50 people including 30 male and 20 female in the stove production and supply chain.

B.6.2. Data and parameters fixed ex ante

SDG13

Data/parameter ID	SDG 13
	ICS 1
Data/parameter	Baseline scenario survey results
Unit	N/A
Description	Report of the results of the baseline scenario survey
Source of data	The report presents the results of the Baseline Scenario Survey, relevant for the baseline scenario definition.
Value(s) applied	The baseline scenario findings are presented in section B.4 of the VPA-DD.
Choice of data or Measurement methods and procedures	The applied data have been informed by the findings of the baseline survey.
Purpose of data	Calculation of baseline emissions
Additional comment	A baseline survey was undertaken as detailed in the VPA-DD section B.4. The surveys were done with 187 participants who were not using the project technology. The survey was undertaken between 20/09/2023 to 29/09/2023 in the three states (Gujarat, Madhya Pradesh, and Uttar Pradesh) where the stoves are being distributed. The data will be applied during the crediting period.

Data/parameter ID	SDG 13																											
	ICS 2																											
Data/parameter	Project technology description																											
Unit	NA																											
Description	The detailed description of the project technology																											
Source of data	The methodology requires that any of the following sources shall be used: - Manufacturer specifications - Certifications by national standards body or an appropriate certification party recognized by national standards body - Commercial guarantee - Technical reports from the installer - For stoves built on-site at the end user location, reports of Standard WBT by stove manufacturer or installer Professional opinion or expert opinion is not accepted as a source for this parameter. <i>The Project technology description has been made based on manufacturer specifications.</i>																											
Value(s) applied	<table><tr><th>Parameters</th><th>Jumbo Stove</th></tr><tr><td>Manufacturer name</td><td>Greenway Grameen Infra pvt</td></tr><tr><td>Size:</td><td>12.4" x 10.6" x 11.6"</td></tr><tr><td>Materials:</td><td>Steel and Aluminium with Bakelite Handles</td></tr><tr><td>Loading Capacity:</td><td>40 kg</td></tr><tr><td>Secondary Air Induction Mechanism:</td><td>Yes</td></tr><tr><td>Warranty:</td><td>2 years</td></tr><tr><td>Fuel Type</td><td>Ergonomic front-loading design</td></tr><tr><td>Operational Lifetime (lifespan)</td><td>72 months (6 years)</td></tr><tr><td>Fuel type</td><td>Fuelwood</td></tr><tr><td>Application/service level</td><td>Domestic cooking</td></tr><tr><td>Energy output</td><td>2.32 kW</td></tr><tr><td>Efficiency</td><td>38%</td></tr></table>		Parameters	Jumbo Stove	Manufacturer name	Greenway Grameen Infra pvt	Size:	12.4" x 10.6" x 11.6"	Materials:	Steel and Aluminium with Bakelite Handles	Loading Capacity:	40 kg	Secondary Air Induction Mechanism:	Yes	Warranty:	2 years	Fuel Type	Ergonomic front-loading design	Operational Lifetime (lifespan)	72 months (6 years)	Fuel type	Fuelwood	Application/service level	Domestic cooking	Energy output	2.32 kW	Efficiency	38%
Parameters	Jumbo Stove																											
Manufacturer name	Greenway Grameen Infra pvt																											
Size:	12.4" x 10.6" x 11.6"																											
Materials:	Steel and Aluminium with Bakelite Handles																											
Loading Capacity:	40 kg																											
Secondary Air Induction Mechanism:	Yes																											
Warranty:	2 years																											
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Fuel type	Fuelwood																											
Application/service level	Domestic cooking																											
Energy output	2.32 kW																											
Efficiency	38%																											
Choice of data or Measurement methods and procedures	Greenway Grameen Infra private is the manufacturer of the project technology, and the manufacturers specifications provide the detailed description of the product technology.																											
Purpose of data	Calculation of baseline emissions																											
Additional comment	N/A																											

Data/parameter ID	SDG 13
	ICS 3
Data/parameter	Expected technical life of project technology
Unit	years
Description	The expected technical life of an individual project technology shall be defined in the VPA-DD.
Source of data	The methodology provides that any of the following sources shall be used: - Manufacturer specifications - Certification by national standards body or an appropriate certification party recognized by national standards body - Commercial guarantee or Guarantee from the installer - For stoves built on-site at the end user location, field reports, which comply with the general requirements for sampling, of average technical life of the same stove type operated under similar conditions (socioeconomic and cultural). Simulation modelling may be applied together with such field reports to estimate the average technical life. Professional opinion or expert opinion is not accepted as a source for this parameter. The PP has sourced the data for the parameter from manufacturer guarantee as per the provided confirmation letter.
Value(s) applied	6
Choice of data or Measurement methods and procedures	Greenway Grameen Infra pvt is the manufacturer of the project technology, and the letter of guarantee serves as a confirmation of the expected technical life of the project technology. Greenway stoves customers have reported unencumbered use of the stoves without any crucial functionality of the stove hindered due to any manufacturing faults. Greenway has stringent quality control checks at their manufacturing facility to ensure that the customers can use the stove with minimal wear and tear affecting the performance and efficiency of the stoves for 6 years in ordinary conditions.
Purpose of data	Calculation of baseline emissions
Additional comment	The expected technical life of project technology is longer than the 5-year crediting period of the project and therefore measures to ensure that end users are provided replacement technology of comparable or higher quality at the end of the technical life, by either replacing with comparable or better technology, or retrofitting essential parts with performance guarantee is not needed.

Data/parameter ID	SDG 3
	ICS 4
Data/parameter	Indoor air pollution (IAP) levels of the project technology
Unit	NA
Description	For projects where cooking will move from outdoor to indoor or where the project technology reduces ventilation (for example, changing from a stove with a chimney to improved stove with no chimney), demonstration that Indoor air pollution (IAP) levels are not worsened in the project scenario compared to the baseline, including PM 2.5 and carbon monoxide (CO) emissions.
Source of data	The methodology provides that for the description of IAP level of the project technology, any of the following sources shall be used: - Certification resulting from a manufacturer's test - Report of field testing of the technology - Report of lab testing of the technology - Results of modelling of the technology's operation under field conditions. - For stoves built on-site at the end user location, existing reports of lab or field testing of similar technology. Professional opinion or expert opinion is not accepted as a source for this parameter. For the IAP level of the baseline scenario, the following sources shall be used: - Certification resulting from of a manufacturer's test, - Report of field testing of the technology, - Report of lab testing of the technology, - Results of modelling of the technology's operation under field conditions, or - Expert opinion For both project and baseline technologies, if none of these are available, reference from published literature or report by independent agencies may be used as evidence, provided it is not more than 5 years old. For the description of IAP level of the project technology,a report of lab testing of the technology is used. For the IAP level of the baseline scenario, reference from published literature reports by independent agencies shall be used as evidence, provided it is not more than 5 years old.
Value(s) applied	Project IAP – WBT test report values:

	Jumbo: 394.49 ¹³ (PM 2.5 mg/MJd) Baseline IAP: 695 ¹⁴ 24-hr µg/m3
Choice of data or Measurement methods and procedures	WBT test report for project IAP and Literature review of published data for baseline IAP
Purpose of data	Calculation of baseline emissions
Additional comment	A study published by the Clean Cooking Alliance in 2016 on "Analysing the costs and benefits of clean and improved cooking solutions (conducted by Sanford School of Public Policy and Duke Global Health Institute) shows that the PM2.5 concentrations from wood burning using traditional stoves was estimated at 695 24-hr µg/m3. Another study done in 2017 on "Comparison of a Traditional Cook Stove with Improved Cook Stoves Based on Their Emission Characteristics", reports a PM2.5 value of 680 ¹⁵ µg/m3+/-128. The first value has been used for purposes of the ex-ante estimation for PM2.5 of the baseline stoves since its more conservative.

¹³ Medium value was 305.97 and CO produced was 4.64g/MJd

¹⁴ [355-1.pdf \(cleancooking.org\)](#) see table 5 on page 19

¹⁵ [\(PDF\) Comparison of a Traditional Cook Stove with Improved Cook Stoves Based on Their Emission Characteristics \(researchgate.net\)](#) see table 1 on page 2

Data/parameter ID	SDG 13 ICS 5
Data/parameter	Avoidance of double counting or double claiming among project participants
Unit	NA
Description	Evidence of avoidance of double counting or double claiming with other parties directly involved with the project or programme.
Source of data	Written assertions with the project developer of the ownership rights and intention of selling the emission reductions resulting from the project activity directed at or signed with all the applicable parties of the following: - Carbon rights waiver signed by end-user transferring carbon rights to Climate Impact Partners (CME)
Value(s) applied	The PP has provided proof that the procedure for the carbon rights transfer with end-users of the project technology is in place.
Choice of data or Measurement methods and procedures	The mechanism for the carbon rights transfer has been put in place and the same shall be verified to ensure there is no double counting or claiming by the project participants.
Purpose of data	Calculation of baseline emissions
Additional comment	N/A

Data/parameter ID	SDG 13 ICS 6
Data/parameter	Avoidance of double counting or double claiming with other mitigation actions
Unit	NA
Description	Review and analysis of mitigation actions in other voluntary market or UNFCCC/compliance mechanisms
Source of data	Using publicly available information from Gold Standard and other voluntary standards, at a minimum Verra and any recognized national or regional standards in the project location, and UNFCCC CDM project & PoA database: - Identify and list any mitigation actions of similar technology, i.e., that provide the same kind of output and use the same kind of equipment or conversion process, operating in overlapping spatial boundaries. If one or more are identified: - Describe the practices that will be implemented to ensure that the programme or project activity quantifies emission reductions only from technology it has implemented, - Describe the practices to avoid that the programme or project activity implementation displaces technology of other mitigation actions, and - Design a method to discount emission reductions in case the programme or project activity is found to displace or operate alongside another mitigation action. The PP has reviewed the Gold Standard and, Verra and UNFCCC CDM project & PoA registries
Value(s) applied	4. PoA GS10818 - Dissemination of Improved Cookstoves in India by Greenway - Dissemination of Improved Cookstoves in Karnataka by Greenway – (VPA001-VPA026) 5. GS11242- Fair Climate Programme for Sustainable Household Energy (PoA) 6. CDM 9181: MicroEnergy Credits – Microfinance for Clean Energy Product Lines – India (unfccc.int)- POA (CPAs 01-37)
Choice of data or Measurement methods and procedures	Data is sourced from the official project registries of the carbon standards.
Purpose of data	Calculation of baseline emissions
Additional comment	Greenway Grameen Infra Private Limited is implementing another Gold Standard Project which uses their project stoves, in addition, the Greenway stoves are also used in another CDM registered project. The following practices will be in put in place to avoid any potential double-counting:

1. In order to ensure that the project activity quantifies emission reductions only from technology it has implemented, each project activity shall maintain a separate and distinct end-user database.
2. Each stove shall be registered under only one project activity and will be clearly distinguished by a unique serial number and cannot be included in any other project.
3. To avoid that the project activity implementation displaces technology of other mitigation actions, a question will be included in the monitoring survey questionnaire to cross-check if the end-users are using other stoves included in other programmes, and the frequency of use.
4. In case the project technology is found to displace or operate alongside other similar stove technologies included in other carbon standards, the stove will be discounted in the emission reductions.

Data/parameter ID	SDG 13 ICS 7
Data/parameter	Regulatory framework for provision of thermal energy services
Unit	NA
Description	Evidence that the project does not undermine or conflict with any national, sub-national or local regulations or guidance for thermal energy supply/devices or fuel supply or use
Source of data	List and provide a summary of any national, sub-national and local regulations or guidance for provision of thermal energy services/devices of the type of the project provides in the project boundary, including any tariff requirements. Describe how the project complies with the regulatory framework.
Value(s) applied	See table 22 below
Choice of data or Measurement methods and procedures	The PP reviewed the regulatory framework for provision of thermal energy services within the host country (India). From the assessment, it has been demonstrated that the project activity does not undermine nor conflict with any regulation or guidance relating to thermal energy supply/devices or fuel supply or use.
Purpose of data	Demonstrate compliance with host country regulations.
Additional comment	N/A

Table 22: Analysis of Relevant Host Country Regulatory Framework

Applicable regulation	Provisions of the law	Project's compliance
Indian Standard for Portable Solid Biomass cookstove (Chulha) [First revision of 13152(Part 1):1991] ¹⁶	The standard provides that the thermal efficiency for each Cookstove when tested in accordance with the method described in Annex A shall not be less than 25 percent for all types of designs and sizes of solid bio-mass natural draft Cookstoves. For forced draft types it shall not be less than 35 percent. The Power Output Rating shall be between 0.5 to 3.0 kWth.	The stove technology promoted by the project activity has been tested and the thermal efficiency achieved for the stoves is 38%. This is above the required minimum of 25% provided in the Standard. The power output for stoves was tested and confirmed to be 2.32 KW which is within the allowed range. The project has therefore complied with the requirements set out.
The National Clean Air Programme (NCAP), 2019 ¹⁷	This is a recent governmental initiative, launched in January 2019 by the Ministry of Environment, Forest and Climate Change (MoEFCC). The Programme focuses on the most polluted Indian cities which are to prepare action plans with time-bound targets to prevent, control, and reduce emissions.	The project activity promotes use of improved biomass stoves which will significantly reduce indoor air pollution among beneficiary households. This will commensurate the government efforts of reducing emissions from cooking and avert related household air pollution deaths and illnesses.
The Digital Personal Data Protection Act, 2023 ¹⁸	An Act to provide for the processing of digital personal data in a manner that recognises both the right of individuals to protect their personal data and the need to process such personal data for lawful purposes. It requires that "A person may process the personal data of a Data Principal only in accordance with the provisions of the Act	The project developer has ensured consent is sought from the end-users when collecting the personal details for purposes of monitoring the stove usage and maintenance of the stoves. Consent shall also be sought during the project surveys. Only end-users voluntarily willing to take part in the survey will be interviewed. The project developer also has an internal data protection

¹⁶ [Final Modified BIS \(Cookstove\) April 2013 MED 04\(1157\)F.pdf \(newdawnengineering.com\)](#)

¹⁷ https://moef.gov.in/wp-content/uploads/2019/05/NCAP_Report.pdf

¹⁸ [Digital Personal Data Protection Act 2023.pdf \(meity.gov.in\)](#)

	and for a lawful purpose, — (a) for which the Data Principal has given her consent; or (b) for certain legitimate uses”.	policy that sets the conduct on how project’s database is handled.
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Data/parameter ID	SDG 13 ICS 8
Data/parameter	$EF_{b,f,CO2}$
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of fuels in baseline scenario
Source of data	Wood: Methodology default, 112 tCO ₂ /TJ
Value(s) applied	112
Choice of data or Measurement methods and procedures	The methodology default value of 112 tCO ₂ /TJ for firewood is applied.
Purpose of data	Calculation of baseline emissions
Additional comment	N/A

Data/parameter ID	SDG 13 ICS 9
Data/parameter	$EF_{b,f,nonCO2}$
Unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor arising from use of fuels in baseline scenario
Source of data	Wood: Methodology default: 9.46 tCO ₂ e/TJ (AR5 GWP)
Value(s) applied	9.46 (AR5 GWP)
Choice of data or Measurement methods and procedures	The methodology AR5 GWP default value for firewood is applied
Purpose of data	Calculation of baseline emissions
Additional comment	N/A

Data/parameter ID	SDG 13
	ICS 12
Data/parameter	$NCV_{b,fuel}$
Unit	TJ/ton
Description	Net calorific value of the fuels used in the baseline
Source of data	Wood: Methodology default, 0.0156 TJ/ton
Value(s) applied	0.0156
Choice of data or Measurement methods and procedures	The methodology default value for firewood is applied
Purpose of data	Calculation of baseline emissions
Additional comment	N/A

Data/parameter ID	SDG 13
	ICS 17
Data/parameter	$fNRB_{I,y}$
Unit	Percentage
Description	Fractional non-renewability status of woody biomass fuel during year y , in case the baseline fuel is biomass or charcoal
Source of data	Determined by following the CDM TOOL30 ¹⁹ , Calculation of the fraction of non-renewable biomass
Value(s) applied	0.9708 (Gujarat) 0.9209 (Madhya Pradesh) 0.9903 (Uttar Pradesh)
Choice of data or Measurement methods and procedures	Calculated
Monitoring frequency	One of two options, with the option defined and fixed at project design certification stage: 1. Determined ex-ante and fixed for a given crediting period or 2. Updated biennially or at each monitoring and verification PP has chosen option 1 and the parameter has been fixed ex-ante for the crediting period
QA/QC procedures	Requirements of the CDM TOOL30
Purpose of data	Calculation of baseline emissions
Additional comment	The project developer conducted a state wise calculation of the fNRB for each of the three states, using the most available forestry data. The data sources and calculations are detailed in the Ex-ante ER sheet.

¹⁹ [EB115_repan22_TOOL30_v04.0 \(unfccc.int\)](https://unfccc.int/eb115_repan22_tool30_v04.0)

Data/parameter ID	SDG13
	ICS 28
Data/parameter	LE _{p,y}
Unit	tCO ₂ e per year
Description	Leakage in project scenario p during year y
Source of data	Methodology default value
Value(s) applied	0.0811 (Gujarat) 0.1133 (Madhya Pradesh) 0.1219 (Uttar Pradesh)
Choice of data or Measurement methods and procedures	Methodology default value is applied
Monitoring frequency	Every two years, or default discount value of 0.95 applied to emission reductions
QA/QC procedures	Compliance with the general requirements for sampling and general requirements for QA/QC
Purpose of data	Calculation of baseline emissions
Additional comment	The methodology default value of 0.95 is applied to account for any leakage and hence no monitoring will be required for the parameter. The parameter value is fixed ex-ante for the crediting period.

B.6.3. Ex ante estimation of SDG Impact

>> **Ex ante calculations related to the outcomes of SDG 13**

Emission reductions

The transparent ex-ante calculations of the outcomes of SDG 13 (emission reductions) are provided in a separate excel spreadsheet uploaded to SustainCert App. For data/parameters available before design certification values contained in section B.6.2 above have been applied and for data/parameters not available before design certification the estimates contained in section B.7.1 below have been used.

The following calculation steps is followed to estimate emission reduction:

$$ER_y = \sum_{b,p} (N_{b,p,y} \times U_{p,y} \times SFS_{p,b,y} \times NCV_{b,fuel} \times (f_{NRB,b,y} \times EF_{b,f,CO_2} + EF_{b,f,nonCO_2})) - \sum LE_{p,y} \quad Eq. 1$$

Ex ante calculations related to the outcomes of SDG 1

Estimated monetary savings computed from fuel saved (Rupee/household/day)

The VPA includes the distribution of fuel-efficient cookstove technologies to replace the inefficient baseline cooking methods and thus result in a reduction in firewood consumption. Using the project technology is expected to lead to financial savings through reduced fuel consumption costs.

The ex-ante estimate for SDG 1 has been calculated based on estimated household savings by assessing the prevailing fuel prices in the host country. This was calculated as follows:

Monetary savings	Value
Average daily expenditure on fuel (baseline)	37.082
Expenditure with project stove	9.76
Savings (Rupee)	27.32

Source: GS 12533 Ex-ante ER spreadsheet tab specific savings

Ex ante calculations related to the outcomes of SDG 5

Average time saved by women for fuel collection (hours/household/day)

The use of a Greenway stove in a household helps in reducing the time spent on firewood collection which is mostly done by women. Fuel collection takes more time and effort, which takes away productive hours of women and young girls leading to time poverty and lost opportunity.

The estimated time saving is calculated as the difference between time spent in the baseline less time spent in the project scenario fuel collection. The time savings has been calculated to be 1.07 hours/household/day. The ex-ante estimation for the parameter has been determined as follows:

Time saving	Value
Average daily time spent in baseline fuel search	1.46
Average daily time spent in project fuel search	0.38
Savings hrs)	1.07

Source: GS 12533 Ex-ante ER spreadsheet tab specific savings

Ex ante calculations related to the outcomes of SDG 7

Number of people/households with access to basic service provided by the project device

The project is expected to distribute 20,000 improved Greenway Jumbo fuelwood cookstoves among household end-users as summarized below.

Ex ante calculations related to the outcomes of SDG 8

Total number of jobs split by gender

The VPA is expected to job creation through the stove production and distribution process. This ex-ante estimation for this parameter has been determined from the estimate provided by the project implementer. Each year the project is expected to employ about 50 people including 30 males, and 20 females.

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 13

Year	Baseline estimate (tCO ₂ e)	Project estimate (tCO ₂ e)	Net benefit (tCO ₂ e)
16/10/2023- 31/12/2023	3,217	0	3,217
2024	40,963	0	40,963
2025	40,851	0	40,851
2026	40,851	0	40,851
2027	40,851	0	40,851
01/01/2028- 15/10/2028	32,345	0	32,345
Total	199,078	0	199,078
Total number of crediting years	5		
Annual average over the crediting period	39,815	0	39,815

Source: GS 12533 Ex-ante ER spreadsheet tab SDGs Summary

SDG 1: Estimated monetary savings computed from fuel saved
(Rupee/household/day)

Year	Baseline estimate	Project estimate (rupee/hh/day)	Net benefit (rupee/hh/day)
16/10/2023-31/12/2023	0.00	6.83	6.83
2024	0.00	27.32	27.32
2025	0.00	27.32	27.32
2026	0.00	27.32	27.32
2027	0.00	27.32	27.32
01/01/2028-15/10/2028	0.00	20.49	20.49
Total	0.00	136.62	136.62
Total number of crediting years	5		
Annual average over the crediting period	0	27.32	27.32

Source: GS 12533 Ex-ante ER spreadsheet tab SDGs Summary

SDG 5: Average time saved by women for fuel collection
(hours/household/day)

Year	Baseline estimate	Project estimate (rupee/year/household)	Net benefit (rupee/year/household)
16/10/2023-31/12/2023	0.00	0.27	0.27
2024	0	1.07	1.07
2025	0	1.07	1.07
2026	0	1.07	1.07
2027	0	1.07	1.07
01/01/2028-15/10/2028	0.00	0.81	0.81
Total	0.00	5.37	5.37
Total number of crediting years	5		
Annual average over the crediting period	0	1.07	1.07

Source: GS 12533 Ex-ante ER spreadsheet tab SDGs Summary

SDG 7: Number of households with access to basic service provided by the project device

Year	Baseline estimate	Project estimate (number)	Net benefit (number)
16/10/2023-31/12/2023	0	20,000	20,000
2024	0	20,000	20,000
2025	0	20,000	20,000
2026	0	20,000	20,000
2027	0	20,000	20,000
01/01/2028-15/10/2028	0	20,000	20,000
Total	0	20,000	20,000
Total number of crediting years	5		
Annual average over the crediting period	0	20,000	20,000

Source: GS 12533 Ex-ante ER spreadsheet tab SDGs Summary

SDG 8: Total number of jobs split by gender

Year	Baseline estimate	Project estimate (number)	Net benefit (number)
16/10/2023-31/12/2023	0	13 people (7males, 6 females)	13 people (7 males, 6 females)
2024	0	50 people (30 males, 20 females)	50 people (30 males, 20 females)
2025	0	50 people (30 males, 20 females)	50 people (30 males, 20 females)
2026	0	50 people (30 males, 20 females)	50 people (30 males, 20 females)
2027	0	50 people (30 males, 20 females)	50 people (30 males, 20 females)
01/01/2028-15/10/2028	0	37 people (23 males, 14 females)	37 people (23 males, 14 females)
Total	0	250 people (150 males, 100 females)	250 people (150 males, 100 females)
Total number of crediting years	5		
Annual average over the crediting period	0	50 people (30 males, 20 females)	50 people (30 males, 20 females)

Source: GS 12533 Ex-ante ER spreadsheet tab SDGs Summary

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data / Parameter ID	SDG 13
	ICS 15
Data / Parameter	Avoidance of double counting or double claiming among project technology end users
Unit	NA
Description	Evidence of avoidance of double counting or double claiming with project technology end-users
Source of data	Carbon title waiver forms signed by end-users transferring their rights to the CME.
Value(s) applied	No double counting. Evidence of carbon waiver transfer to be provided during verification.
Measurement methods and procedures	NA
Monitoring frequency	Monitored whenever project technology is sold.
QA/QC procedures	Cross check using general internet search and search of public records of Gold Standard and other voluntary market and UNFCCC mechanisms
Purpose of data	Calculation of emission reductions
Additional comment	N/A

Data / Parameter ID	SDG 13 ICS 16
Data / Parameter	Presence of stove stacking
Unit	NA
Description	Descriptive statistics of the presence and usage practices of baseline- and other non-project-technology by project technology end users.
Source of data	Usage survey The PP will use usage Survey to assess this parameter
Value(s) applied	33.1% (Gujarat) 1.9% (MP) 1.3% (UP)
Measurement methods and procedures	Surveys
Monitoring frequency	Annual
QA/QC procedures	The calculation of $SFS_{p,b,y}$ shall be cross-checked with the observed presence of stove stacking. Ensure any stove stacking is considered so that emission reductions are calculated only from real reduction of, or replacement of, baseline fuel use.
Purpose of data	Calculation of emission reductions
Additional comment	The values applied are based on the LPG connection assessment done during the installation survey. Actual values to be determined during the usage survey.

Data / Parameter ID	SDG 13 ICS 18
Data / Parameter	$P_{b,y}$
Unit	kg/household-day
Description	Quantity of fuel that is consumed in baseline scenario b during year y
Source of data	Kitchen Performance Tests
Value(s) applied	7.425
Measurement methods and procedures	Measured
Monitoring frequency	At the start of crediting period (fixed for one crediting period)
QA/QC procedures	Compliance with the general requirements for sampling (Section 4.4), general requirements for QA/QC (Section 4.5) and Annex 2 Kitchen performance test.
Purpose of data	Calculation of emission reductions
Additional comment	Used to calculate SFS under method 1. The methodology sets a threshold value of 0.75 tonnes/person/year. The PP can confirm that the

project has not exceeded this limit as the baseline fuel consumed per household per year is 2.71 tonnes/household/year which translates to 0.543 tonnes/person/year (average household size calculated at 4.98).

Data / Parameter ID	SDG 13 ICS 19
Data / Parameter	$P_{p,y}$
Unit	kg/household-day
Description	Quantity of fuel that is consumed in project scenario p during year y
Source of data	Project performance field tests
Value(s) applied	3.423
Measurement methods and procedures	Measured
Monitoring frequency	Updated every two years, or more frequently
QA/QC procedures	Compliance with the general requirements for sampling (Section 4.4), general requirements for QA/QC (Section 4.5) and Annex 2 Kitchen performance test.
Purpose of data	Calculation of emission reductions
Additional comment	Used to calculate SFS under method 1.

Data / Parameter ID	SDG 13 ICS 20
Data / Parameter	$SFS_{b,p,y}$
Unit	kgs/household-day
Description	Specific fuel savings for an individual project technology in baseline b/project p pair in year y
Source of data	Calculated from $P_{b,y}$, $P_{p,y}$ and other information to obtain the savings in the required units
Value(s) applied	4.00
Measurement methods and procedures	Calculated
Monitoring frequency	Updated every two years, or more frequently
QA/QC procedures	The calculation method, inputs and their sources shall be described in detail in the VPA-DD and monitoring report.
Purpose of data	Calculation of emission reductions
Additional comment	The value applied has been determined from the KPT completed by the project. The baseline and project field test data has been analyzed in combination to estimate the average fuel savings per technology unit. This was calculated from $P_{b,y}$ (7.42) – $P_{p,y}$ (3.42)= 4 kg/hh/day.

Data / Parameter ID	SDG 13 ICS 26
Data / Parameter	$U_{p,y}$
Unit	Percentage
Description	Weighted average usage rate in project scenario p during year y
Source of data	Usage survey exercise
Value(s) applied	90
Measurement methods and procedures	Calculated. The survey result shall provide the statistically valid proportion of users actively using the project technology for each project technology age cohort. From the annual usage survey results, the weighted average percent of users actively using the project technology will be calculated, where the weighting is by the quantity of project technologies of each age cohort being credited in a given project scenario.
Monitoring frequency	Annually
QA/QC procedures	Compliance with the general requirements for sampling (Section 4.4) and general requirements for QA/QC (Section 4.5)
Purpose of data	Calculation of emission reductions
Additional comment	The value applied is an estimate for purposes of ex-ante emission calculations. Actual value to be determined during monitoring.

TEMPLATE

Data / Parameter ID	SDG 13 ICS 27
Data / Parameter	$N_{b,p,y}$
Unit	days
Description	Number of project technology-days included in the project database for baseline b/project p pair in year y
Source of data	Calculated from the Project database as the sum of the number of project technology units times the calendar days during the year that they were present at the end user locations
Value(s) applied	365
Measurement methods and procedures	Calculated
Monitoring frequency	Annually
QA/QC procedures	Cross check the results of the usage survey with the contents of the project database to confirm whether the project technology units surveyed are present at end user locations as expected, or not.
Purpose of data	Calculation of emission reductions
Additional comment	The value applied is an estimate for purposes of ex-ante calculation, actual value to be determined during monitoring.

SDG 1

Data / Parameter	SDG 1 No poverty
Unit	Rupee/household/day
Description	Estimated monetary savings computed from fuel saved.
Source of data	Project surveys
Value(s) applied	27.32
Measurement methods and procedures	Surveys
Monitoring frequency	Annual
QA/QC procedures	Questions will be included in the survey questionnaire to assess the parameter. Survey results will determine the proportion of population who demonstrate achieving monetary saving from use of the project technology.
Purpose of data	Reporting on SDG impacts
Additional comment	The value applied is an estimate for purposes of ex-ante calculation, actual value to be determined during monitoring.

SDG 5

Data / Parameter	SDG 5 Gender Equality
Unit	hours/household/day
Description	Average time saved by women from fuel collection.
Source of data	Project records and surveys
Value(s) applied	1.07
Measurement methods and procedures	Surveys
Monitoring frequency	Annual
QA/QC procedures	Questions will be included in the survey questionnaire to assess the parameter. Survey results will determine the average time saving associated with fuel collection among end-users.
Purpose of data	Reporting on SDG impacts
Additional comment	The value applied is for purposes of ex-ante estimation. Actual value to be determined during monitoring.

SDG 7

Data / Parameter	SDG 7 Access to clean energy
Unit	Number
Description	Number of people/households with access to basic service provided by the project device
Source of data	Project sales database
Value(s) applied	20,000
Measurement methods and procedures	The sales records will be checked to determine the number of people using the project technology.
Monitoring frequency	Annual
QA/QC procedures	An updated sales database shall be maintained by the project implementor.
Purpose of data	Reporting on SDG impacts
Additional comment	Actual number will be determined during monitoring.

SDG 8

Data / Parameter	SDG 8 Decent Work and economic growth
Unit	Number of people
Description	Total number of jobs split by gender
Source of data	Employment records
Value(s) applied	50 (30 male, 20 female)
Measurement methods and procedures	To be determined from employment records
Monitoring frequency	Annual
QA/QC procedures	The project implementor shall maintain up to date employment records to confirm the number of people employed per department, split by gender.
Purpose of data	Reporting on SDG impacts
Additional comment	The value applied is the projected number. Actual number will be determined during monitoring.

Safeguards Monitoring

Data / Parameter	Principle P 3.1.2
Unit	Number of people
Description	Number of workers trained on health and safety
Source of data	Training records
Value(s) applied	50
Measurement methods and procedures	To be determined from training records
Monitoring frequency	Annual
QA/QC procedures	The project implementor shall maintain up to date training records to confirm the number of workers trained of health and safety.
Purpose of data	Reporting on safeguards monitoring.
Additional comment	The value applied is the projected number. Actual number will be determined during monitoring.

B.7.2. Sampling plan

>> **Sampling Objective and Parameter to be monitored**

Section 4.4. of the applicable methodology outlines the general requirements for sampling and provides that when sampling is applied to determine mean (average) parameter values or proportion (yes/no) parameter values for both ex-ante and monitored data and parameters, the outlined guidelines shall always be applied. Furthermore, specific requirements apply for the sampling related to some parameters.

This project will apply Simple Random Sampling. This will be done in accordance with the Guideline for Sampling and surveys for CDM project activities and programme of activities, version 4. The project developer shall maintain an updated distribution record for all the units sold, and this will identify the end-users and their locations within the project activity. This will enable random sampling and testing of subjects to be identified from the distribution records.

The parameters to be monitored by the project activity will be the fuel consumption trends, drop-off rates and project usage rates. The target population monitored parameters (proportion of stoves in operation ($U_{p,y}$), presence of stove stacking and fuel consumption and savings ($SFSp,b,y$)) for this project shall be all the households within the project activity database using the project stoves. The determination of the presence of stove stacking will be conducted together with the usage survey.

As outlined in section 4.1.8 of the applicable methodology, the objective of the usage survey will be to provide a single usage parameter that is weighted based on the age distribution for project technologies in the Project Database, Up,y . This usage parameter will be established to account for drop off rates as project technologies age and are replaced, or for reduced use by end users for other motives.

The usage survey shall determine the usage proportion for each age cohort of technologies being credited for each project scenario p. The age cohorts in the survey will be established as follows:

- Participants in a usage survey with technologies in the first year of use (age0-1) must have technologies that have been in use on average at least 0.5 years or longer.

- Participants in a usage survey with technologies in the second year of use (age1-2) must be conducted with technologies that have been in use on average at least 1.5 years, and so on.

The sample size shall be defined for each age cohort following the general requirements for sampling outlined in the applied methodology with a minimum of 30 samples for project technologies of each age cohort being credited, except where the age cohort comprises fewer than 30 units, in which case all units shall be sampled.

Target population

The target population will be the total stove population in the project activity.

Sampling Methodology

Since the baseline cooking patterns and behaviours are the same across the project boundary, the usage pattern is expected to be similar. The project will employ a simple random sampling method when drawing up a sample.

Sample Size

The minimum sample sizes for the determination of the monitored parameters will be determined as per 'The Guidelines for Sampling and Surveys for CDM Project Activities and Programme of Activities', Version 04.0 and the registered PDD. The minimum sample sizes required for the determination of the different types of parameters (proportion and mean) will be determined using different methods as described below.

1. Usage survey sample size calculation (proportion parameter), presence of stove stacking and Specific fuel savings for an individual technology of project (mean parameter)

When determining the values of interest i.e. proportion parameter and mean parameter, the following equation for the sample size will be used to calculate sample size when applying stratified random sampling (As provided in the Guidelines for Sampling and Surveys for CDM Project Activities and Programme of Activities', Version 04.0). The usage proportion for each age cohort of technologies being credited shall be determined following the criteria discussed above. A minimum of 30 samples for project technologies of each age cohort shall be maintained, except where the age cohort comprises fewer than 30 units, in which case all units shall be sampled.

Proportion parameter

$$n \geq \frac{1.645^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.645^2 p(1-p)} \quad \text{Equation (1)}$$

Mean parameter.

$$n \geq \frac{1.645^2 NV}{(N-1) \times 0.1^2 + 1.645^2 V} \quad \text{Equation (18)}$$

Baseline and project survey

When carrying out baseline and project surveys, the following sample size guidelines should be applied, unless otherwise stated for specific parameters:

- Group size <300: Minimum sample size 30 or population size, whichever is smaller.
- Group size 300 to 1000: Minimum sample size 10% of group size
- Group size > 1000 Minimum sample size 100

General Requirements for Quality Assurance and Quality Control

The project developer is responsible for accurate and transparent record keeping, monitoring and evaluation. All supporting documentation and records for the project must be easily accessible for spot checking and cross referencing by a third-party auditor.

Contact information in the total sales/distribution record shall allow a project auditor to easily contact and visit end users.

An auditor must be able to cross reference pertinent project documentation, including archives such as production records (e.g., materials purchases, internal logs), financial accounts and sales records, as well as wholesale customer invoices, observations of retailer activities and sales performance.

B.7.3. Other elements of monitoring plan

>> The following shall be monitored and recorded during project crediting period.

a) Total sales or dissemination record

The project developer shall maintain an accurate and complete “dissemination record” or “installation record.” The record shall be maintained continuously and backed up electronically. The required data for each project unit is:

1. Date of installation
2. Geographic area of sale
3. Model/type of project technology sold
4. Quantity of project technologies sold
5. Name, telephone number (if available), and address and/or GPS coordinates: a. Required for all bulk purchasers, i.e., retailers and industrial users; b. Required for all end users, e.g. households
6. Mode of use: domestic, institutional commercial, other.

b) Project Database

The project database lists all the project technology units that have been distributed by the project and have not surpassed their technical life. It will be derived from the total dissemination record and shall be maintained continuously. Within the database, project technologies units are labelled, at a minimum, with their corresponding project scenario p and their date of dissemination. Technologies aged beyond their technical life, and not replaced or retrofitted, are removed from the project database, and no longer credited.

c) Project Surveys

The project surveys include the usage survey ($U_{p,y}$), Usage Survey- use of other stoves (ICS 16), and the baseline and project performance field tests (BFT and PFT).. The project surveys have the same sample sizing and data collection guidelines as the baseline scenario survey. The project surveys shall only be conducted with the end user’s representative of the project scenario and currently using the project technology.

i) Usage Survey ($U_{p,y}$)

The objective of the usage survey is to provide a single usage parameter that is weighted based on the age distribution for project technologies in the Project Database, $U_{p,y}$. This usage parameter must be established to account for drop off

rates as project technologies age and are replaced, or for reduced use by end users for other motives.

The usage survey shall determine the usage proportion for each project technology being credited for each project scenario p . The usage survey shall be conducted in line with the requirements and guidelines of sampling.

ii) Usage Survey- use of other stoves

As part of the usage survey, the project developer shall collect data on the presence and usage practices of baseline and other non-project technology by project technology end users and prepare descriptive statistics of these practices. The same method of in person interviews and expert observation within the kitchen in question will be used to collect these data. Where the usage survey indicates usage of other non-project technology, this shall be addressed in line with the requirements for parameter ICS 20.

d) Baseline Fuel Consumption

To calculate SFS under method 1 applied, the project developer shall determine $P_{b,y}$, Quantity of fuel that is consumed in baseline scenario b during year y in units of kg/household-day, kg/person-meal, etc. The baseline fuel consumption shall be determined using project specific Baseline Performance Field Tests (BFT). The BFT will be completed once ex-ante, and at crediting period renewal.

e) Project Fuel Consumption

To calculate SFS under method 1 applied, the project developer shall determine $P_{p,y}$, Quantity of fuel that is consumed in project scenario p during year y in units of kg/household-day, kg/person-meal, etc. The project fuel consumption shall be determined using a project specific Project Performance Field Test (PFT). The PFT is completed every other year, or more frequently, and the first PFT shall be completed before the first verification of the project.

f) Data Archiving

Data collected for the purposes of monitoring the VPA will be archived in electronic format and where possible in hard copies. The same will be availed to VVB during verification when required.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of VPA

>>16/10/2023 (date when the first stove was sold). The project is a retroactive project since the first LSC was conducted on 10/11/2023 which is before the commencement of the stove's distribution.

C.1.2. Expected operational lifetime of VPA

>>15 years 00 Months

C.2. Crediting period of project

C.2.1. Start date of crediting period

>>16/10/2023 ²⁰(start of the project)

C.2.2. Total length of crediting period

>>5 years 00 Months (renewable twice)

²⁰ As per the GHG requirements para 10.2.1, the start date of Crediting Period is the date of start of operation or a maximum of two years prior to the date of Project Design Certification, whichever occurs later.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

PRINCIPLES	MITIGATION MEASURES ADDED TO THE MONITORING PLAN
Principle 3.1.2	Workers’ training on health and safety.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project has a policy which requires non-discrimination of employees and put measures in place for equal treatment to all its workers. The non-discrimination covered by the policy encompasses discrimination based on gender. This policy will continue to be enforced and appropriate measures will be undertaken to ensure that the same is complied with. The policy provides for means of addressing discrimination whenever it arises.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The project developer abides by the principle of gender equality as enshrined in the Indian Constitution in its Preamble, Fundamental Rights, Fundamental Duties and Directive Principles. The Constitution grants equality to women. The project will, therefore, aim to abide by the constitution and any applicable laws.

Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?

No

Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?

No

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The GS requirement is that projects shall carry out two rounds of consultations. The requirement stipulates that at least one of the consultations should be a physical meeting while the other can be physical or be done online.

Physical meeting

The project implementer conducted two physical stakeholders' consultation meetings which were held on 10/11/2023 and 20/11/2023.

Stakeholder Feedback Round (SFR)

The SFR was run online where the stakeholders who were invited for the physical stakeholder meeting (first consultation) were invited to participate in the second stakeholder feedback round.

The SFR process was conducted for a period of 60 days between 24/01/2024 to 24/03/2024. The documents sent out to the stakeholders included the stakeholder's consultation report, the non-technical summary and stakeholder feedback form. The stakeholders were requested to review the documents and provide comments before the expiry of the 60-day period.

No feedback was received from the SFR consultation.

Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

>>

Gender of Stakeholder	Stakeholder comment	Was comment taken into account (Yes/No)?	Explanation/ Justification (Why? How?)
Female	How safe is the Greenway cookstove?	Yes	The cookstoves have been created after many stages of testing and are completely safe to use.
Male	How will the product reach all the beneficiaries to whom its use could be of great help?	Yes	Greenway will be conducting many awareness campaigns, along with its partner organizations in various talukas and districts of Karnataka. These will help the product to reach more users who can benefit greatly with the help of Greenway's cookstoves.
Female	Can you explain how I am contributing to climate change?	Yes	The concept of climate change is a long-term shift in the average weather conditions of a region, such as its average temperature, and rainfall due to release of certain gases in the atmosphere. We all participate in some form or the other by using cars, or taking flights, etc. In villages, one of the main contributors to climate change is the smoke produced while using traditional cookstoves.

E.2. Final continuous input / grievance mechanism

METHOD	INCLUDE ALL DETAILS OF CHOSEN METHOD (S) SO THAT THEY MAY BE UNDERSTOOD AND, WHERE RELEVANT, USED BY READERS.
Continuous Input / Grievance Expression Process Book (mandatory)	The Grievance Redressal Mechanism book is placed in these locations: Location 1: Avichal Ventures, Khasra No 224, Village Oriya Tehsil Panagar, Katangi Bypass, Mangela, Jabalpur, Madhya Pradesh – 482002. Location 2: Greenway Grameen Infra Pvt. Ltd., 23/A, GIDC Manjusar, Taluka Savli, Vadodara – 391775
GS Contact (mandatory)	help@goldstandard.org
Telephone access (optional)	(+91) 7861098770
Internet/email access (optional)	contact@greenwayappliances.com
Nominated Independent Mediator (optional)	N/A
Other	

SECTION F. Eligibility and inclusion criteria for VPAs inclusion

>>

The below table shall be completed for all VPAs.

The CME shall provide clear description on how eligibility criteria set at real case VPAs are complied with for each real case and regular VPAs submitted for inclusion.

The CME shall not change the eligibility criteria and required condition set at real case VPAs. At the time of inclusion of regular VPAs, the CME shall only describe how the regular VPAs comply with the eligibility criterion.

NO.	ELIGIBILITY CRITERION	DESCRIPTION/ REQUIRED CONDITION	DESCRIPTION OF THE VPA IN RELATION TO THE CRITERIA, MEANS OF VERIFICATION AND SUPPORTING EVIDENCE FOR INCLUSION
1.	<i>Geographical boundaries of V/CPAs must be consistent with the geographical boundary of the PoA;</i>	The VPA shall be located within the PoA boundary of India	The VPA is being implemented in the states of Gujarat, Madhya Pradesh and Uttar Pradesh which are within the Republic of India (the PoA boundary).
2.	<i>Conditions to avoid double counting of Impacts, such as unique identifications of product and end user locations (e.g. programme logo);</i>	Identification of each device through unique identification number. Each VPA will show that it is exclusive to the PoA and not registered as another project activity or VPA under another PoA.	Greenway stoves under this project activity have a unique serial number that ensures they are uniquely identifiable to this project. Further, each end user shall have only one registered project stove to ensure there is no double counting. An updated sales database will be maintained and will record the number of stoves sold to each end-user. Only one stove shall be eligible for crediting. The PP has also signed a letter confirming that

			the VPA is only registered to the PoA.
3.	<i>Conditions to confirm that V/CPAs are neither registered as project activities with other offset Schemes, included in other registered PoAs, nor the project activities that have been deregistered;</i>	Confirmation that the VPA is neither registered with other carbon standards	The CME has declared that the VPA is neither registered as a project activity with GS or any other standard or as a VPA of another PoA. The PP conducted a search on the Gold Standard, CDM and VERRA registries to check the following: (i) Registration of the project with other standards (ii) similar projects registered within the same project boundary (iii) projects using the same project technology as the project and (iv) whether the project activity had been deregistered by any of the standards. The search was conducted on 23/04/2024 and the PP can confirm that the VPA is not registered with other Carbon Standards. However, from the search it was established that there are other programs that were found to be making use of the Greenway stoves as follows: 7. PoA GS10818 - Dissemination of Improved Cookstoves in India by Greenway - Dissemination of Improved Cookstoves in Karnataka by Greenway – (VPA001-VPA026) 8. GS11242- Fair Climate Programme for Sustainable

			Household Energy (PoA) 9. CDM 9181: MicroEnergy Credits – Microfinance for Clean Energy Product Lines – India (unfccc.int)- POA (CPAs 01-37) To ensure there is no double counting, Greenway have a robust stove identification system where each stove in each project is clearly distinguished by a unique serial number and cannot be included in any other project. The projects are implemented by different project developers and each project maintains a separate and distinct database hence avoiding any risk of double counting. The CME can also confirm that the VPA is not a deregistered VPA which is being introduced again. Although there are other projects whose boundaries overlap with the proposed project, the CME confirms that appropriate procedures have been put in place to distinguish each project database and separate each unique stove serial number being included in their respective project databases and this action prevents double counting other mitigation outcomes
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			(See ICS 6 for more details).
4.	<i>Specification of the technology/measure such as the level and type of service, as well as performance specification based on, inter alia, testing/certification;</i>	The VPA shall involve the distribution and sale of highly efficient biomass burning stoves with 20% efficiency and above. Each VPA will involve the distribution and installation of efficient cookstoves to households and/or communities currently cooking with non-renewable biomass.	As per the lab test results provided by the PP, the efficiency of the Greenway Jumbo stoves was tested and has been confirmed as 38%. The project sells stoves to households only.
5.	<i>Conditions to check the start dates of V/CPAs through documentary evidence;</i>	The start date of the VPA shall not be before the PoA stakeholders' consultation date.	The PoA design consultation was initiated on 14/10/2020. The first stakeholder consultation meeting for the VPA was held on 10/11/2023. The VPA start date is 16/10/2023 which marks the date when the first stove was sold under the VPA. The stove database shows the actual distribution dates for the stoves and PP has also submitted supporting carbon waiver signed by the end-user as proof of start date (see Evidence of Start date) and the provided stakeholder's consultation report which shows the date when the consultations were done.
6.	<i>Conditions to ensure compliance with the applicability of the applied methodologies, the applied standardized baselines and the other applied methodological regulatory documents</i>	The VPA shall meet the applicability requirements of the methodology "Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version 4.0" and GS4GG Principles and Requirements.	The VPA applies an approved Gold Standard methodology "Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version 4.0". The applied methodology is applicable to the VPA since the methodology requires that only activities that introduce technologies and/or

			practices that and non-domestic premises can apply it. Since the project introduces stoves technologies that reduce greenhouse gas (GHG) emissions from the thermal energy consumption of households, the VPA therefore meets this condition.
7.	<i>Conditions to ensure that V/CPAs meet the requirements for demonstration of additionality</i>	Each VPA within the PoA shall demonstrate additionality through the Gold Standard additionality criteria or the latest version of the CDM Methodological Tool for the demonstration and assessment of additionality.	Assessment of additionality has been done at VPA level as outlined in section B.5 of the VPA-DD. Documentation: VPA-DD
8.	<i>Conditions to ensure no diversion of official development assistance;</i>	Confirmation of non-involvement of public funding or ODA from Annex I Parties in VPA	The Project activity have signed and ODA declaration form. The signed form has been made uploaded on SustainCert app.
9.	<i>Target group (e.g. domestic/commercial/industrial, rural/urban, grid connected/offgrid), and where applicable, distribution mechanisms (e.g. direct installation)</i>	Target groups should be identified	The VPA distributes improved fuelwood cooking devices to domestic end users and the distribution mechanism is outlined in section A.1 of the VPA-DD.
10.	<i>Conditions related to sampling requirements for the PoA</i>	Sampling requirements be outlined in section B.2 of the PoA shall be applied for the VPAs included into the programme.	The VPA will apply the latest Guidelines for sampling and surveys for CDM project activities and programmes of activities and the sampling process has been outlined in section B.7.2 of the VPA-DD.
11.	<i>Conditions to confirm that technologies in V/CPAs are eligible</i>	All VPAs to meet eligibility requirements specified in section B.3 of the PoA-DD whereby the stoves utilise non-renewable biomass as feedstock and will use fuelwood as the fuel type	The VPA is installing the improved Greenway Jumbo cookstove whose rated thermal efficiency is 38%. This is above the minimum efficiency of 20% required under the methodology. The

			details of the stove have been explained in section A.3 of this VPA-DD. The fuel being used is non-renewable biomass feedstock and fuelwood as the only fuel type.
12.	<i>Conditions to be met by each VPA regarding SDG outcomes assessment</i>	Each VPA shall conduct an SDG Outcomes assessment and shall positively contribute to the sustainable development goals identified in the PoA-DD.	An SDG outcomes assessment has been done in the VPA-DD in section B.6
13.	<i>Conditions to be met by each VPA regarding safeguarding principles</i>	Each VPA shall conduct a safeguarding principles assessment in line with the Gold Standard Safeguarding principles assessment requirements.	A safeguarding principles assessment has been done in the VPA-DD as detailed in Appendix 1
14.	<i>Legal Ownership</i>	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated.	End users shall sign a Carbon Waiver Form during purchase which transfers the rights of the carbon credits to the CME (Climate Impact Partners).
15.	<i>On-going financial need</i>	<i>Each VPA to demonstrate on-going financial need</i>	As per para 4.1.52 of Principles and Requirements, Ongoing Financial Need shall be demonstrated at Design Certification Renewal.

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

SOCIAL SAFEGUARDING PRINCIPLES		
Reference requirement	Question	Response
P.1 HUMAN RIGHTS		
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project developer, its representatives and the Project disrespect internationally proclaimed human rights?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Is the project involved or complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Have local communities or individuals raised human rights concerns regarding the project (e.g., during the stakeholder engagement process, grievance processes, public statements)?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Is there a risk that rights-holders (e.g., Project-affected stakeholders) do not have the capacity to claim their rights?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does this project undermine national or regional measures for the realisation of the right to development?	☐ YES ☒ NO
If the answer to any of the questions above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements. <i>The constitution of the host country (India) considers it a legal offence to violate human rights during any business activity. India endorses the United Nations Guiding Principles (UNGPs) on Business and Human Rights adopted in the UN Human Rights Council (UNHRC) in 2011. The CME complies with the legal requirements of the host country. This is not violated at any point during the project.</i> Would the project potentially involve or lead to:		

ERROR! REFERENCE SOURCE NOT FOUND.	adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalised groups?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	inequitable or discriminatory impacts on affected populations, particularly people living in poverty or marginalised or excluded individuals or groups, including persons with disabilities?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	restrictions in availability, quality of and/or access to resources or basic services, in particular to marginalised individuals or groups, including persons with disabilities?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	exacerbation of conflicts among and/or the risk of violence to project-affected communities and individuals?	☐ YES ☐ POTENTIALLY ☒ NO

Briefly describe below how the project incorporates a human rights-based approach.

For example, by describing how the project design:

- is informed by human rights analysis, including from UN human rights mechanisms (human rights treaty bodies, universal periodic review, special procedures)
- includes measures to assist the government to realise (respect, protect and fulfil) human rights under international law and to implement human rights-related standards in national law (whichever is higher)
- enhances the availability, accessibility and quality of benefits and services for potentially marginalised individuals and groups, and to increase their inclusion in decision-making processes that may impact them (consistent with the non-discrimination and equality human rights principle)
- provides reasonable accommodations to strengthen inclusivity and accessibility of project benefits and services to persons with disabilities.

The project activity entails distribution of efficient cookstoves to households within the states of Gujarat, Madhya Pradesh, and Uttar Pradesh. This will enhance access to efficient cooking solutions to beneficiary households. All individuals including marginalized individuals and groups will have an equal opportunity to access the cookstoves.

P.2 | GENDER EQUALITY AND WOMEN'S EMPOWERMENT

ERROR! REFERENCE SOURCE NOT FOUND.	Have women's groups/leaders raised gender equality concerns regarding the project, (e.g., during the stakeholder engagement process, grievance processes, public statements)?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project undermine the principles of non-discrimination, equal treatment, and equal pay for equal work?	☐ YES ☒ NO

ERROR! REFERENCE SOURCE NOT FOUND.	Does the project prevent men and women from having equal opportunities to participate in identified tasks and activities, whether through paid work, volunteer work, or community contributions, as appropriate?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project limit the participation of women or men based on pregnancy, maternity/paternity leave, or marital status?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Is information about project objectives being communicated in a way that is inappropriate for the local context and not tailored to the methods of understanding of both women and men, which could hinder their participation?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Has the project assessed gender risks without referencing the country's gender strategy or equivalent national commitment?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Has expert stakeholder(s) been involved, and has their input been requested for the project design on gender equality and women's empowerment?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements.

The project enhances women's access and entitlement of benefits. Since the women will be direct user of the stoves, it will benefit women by reducing their exposure to the indoor air pollution thereby improving their health. in addition, the decrease in quantity of firewood required will reduce workload of women for the collection of firewood. reduced workload for firewood collection results in time saving that the women can use for other productive activities. the project will contribute the above-mentioned benefits, regardless of the marital status of women.

Would the project potentially involve or lead to:

ERROR! REFERENCE SOURCE NOT FOUND.	adverse impacts on gender equality and/or the situation of women and girls?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	exacerbation of risks of gender-based violence? For example, through the influx of workers to a community, changes in community and household power dynamics, increased exposure to unsafe public places and/or transport, etc.	☐ YES ☐ POTENTIALLY ☒ NO

ERROR! REFERENCE SOURCE NOT FOUND.	reproducing discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	limitations on women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well-being.	☐ YES ☐ POTENTIALLY ☒ NO

Briefly describe below how the project is addressing any identified risk to gender equality and women's empowerment.

Households' duties related to firewood collection and cooking predominantly remain with women. The project seeks to decrease burden on women and would not result in social isolation of men. This will have positive impact on women due to reduced time in fuel collection and cooking. This time saving can be utilized by women to engage in other income generating activities. Women will also have equal access to arising job opportunities from the project activity.

In addition, the implementation of the project will contribute towards saving of common resources in form of "firewood" due to fuel saving.

The project duly accounts for the gender roles. Time saving is one of the key benefits from the project which the beneficiary can utilize to fulfil their gender roles. With the saved time, one can perform the respective role more effectively.

P.3 | COMMUNITY HEALTH AND SAFETY

ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve potential risks to the health and safety of affected communities during its life cycle?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve any potential risks to the workers' safety and health?	☒ YES ☐ NO

If the answer to any of the questions above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements.

The project activity entails manufacture and distribution of Greenway stoves among household end-users. The manufacture of the stoves is done in a modern facility and uses efficient technology to produce the stoves. The factory where the stoves are produced has been designed to maximise on the health, safety and working conditions for the workers. However, like other manufacturing processes, there is potential health and safety related risks for the factory workers. The project implementer will be required to provide prerequisite training for the workers to safeguard their health, safety, and general well-being. There are no anticipated risks to the wider community as a result of implementation of the project.

This principle will be monitored to ensure workers are being trained as needed.

Would the project potentially involve or lead to:

ERROR! REFERENCE SOURCE NOT FOUND.	construction and/or infrastructure development (e.g., roads, buildings, dams)?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	air pollution, noise, vibration, traffic, injuries, physical hazards, poor surface water quality due to runoff, erosion, sanitation?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	harm or losses due to failure of structural elements of the project (e.g., collapse of buildings or infrastructure)?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	risks of water-borne or other vector-borne diseases (e.g., temporary breeding habitats), communicable and noncommunicable diseases, nutritional disorders, mental health?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g., explosives, fuel and other chemicals during construction and operation)?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	adverse impacts on ecosystems and ecosystem services relevant to communities' health (e.g., food, surface water purification, natural buffers from flooding)?	☐ YES ☐ POTENTIALLY ☒ NO

Briefly describe below how the project is addressing any identified risk related to community health and safety.

The project activity entails distribution of Greenway stoves among household end-users and is not anticipated to have any impact on the above principles.

P.4 | CULTURAL HERITAGE, INDIGENOUS PEOPLE, DISPLACEMENT AND RESETTLEMENT

P.4.1 | SITES OF CULTURAL AND HISTORICAL HERITAGE

P.4.1.1 	Does the project involve altering, damaging, or removing sites, objects, or structures of significant cultural heritage?	☐ YES ☒ NO
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If the answer to question above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements.

The project activity entails distribution of Greenway stoves for cooking among households and is not anticipated to have any impact on cultural heritage and indigenous people.

Would the project potentially involve or lead to:		
<u>ERROR!</u> <u>REFERENCE</u> <u>SOURCE</u> <u>NOT</u> <u>FOUND.</u>	activities adjacent to or within a cultural heritage site?	☐ YES ☐ POTENTIALLY ☒ NO
<u>ERROR!</u> <u>REFERENCE</u> <u>SOURCE</u> <u>NOT</u> <u>FOUND.</u>	significant excavations, demolitions, movement of earth, flooding or other environmental changes?	☐ YES ☐ POTENTIALLY ☒ NO
<u>ERROR!</u> <u>REFERENCE</u> <u>SOURCE</u> <u>NOT</u> <u>FOUND.</u>	alterations to landscapes and natural features with cultural significance?	☐ YES ☐ POTENTIALLY ☒ NO
<u>ERROR!</u> <u>REFERENCE</u> <u>SOURCE</u> <u>NOT</u> <u>FOUND.</u>	adverse impacts to sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g., knowledge, innovations, practices)? (Note: projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	☐ YES ☐ POTENTIALLY ☒ NO
<u>P.4.1.2</u>	Utilisation of tangible and/or intangible forms (e.g., practices, traditional knowledge) of Cultural Heritage for commercial or other purposes?	☐ YES ☐ POTENTIALLY ☒ NO
<u>ERROR!</u> <u>REFERENCE</u> <u>SOURCE</u> <u>NOT</u> <u>FOUND.</u>	If answer to question above is "YES" or "POTENTIALLY" – are the communities made aware of their right under the law, scope and nature of proposed development and its potential consequences?	☐ YES ☐ NO ☒ NA
<u>ERROR!</u> <u>REFERENCE</u> <u>SOURCE</u> <u>NOT</u> <u>FOUND.</u>	If answer to question above is "YES" – does the project provide equitable sharing of benefits from commercialisation of such knowledge, innovation, or practice, consistent with their customs and traditions?	☐ YES ☐ NO ☒ NA
<u>ERROR!</u> <u>REFERENCE</u> <u>SOURCE</u> <u>NOT</u> <u>FOUND.</u>	If answer to question above is "YES" – are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA
<u>ERROR!</u> <u>REFERENCE</u> <u>SOURCE</u> <u>NOT</u> <u>FOUND.</u>	If answer to question above is "YES", has project design been changed, modified, updated considering opinions and recommendations of an Expert Stakeholder?	☐ YES ☐ NO ☒ NA

If the answer is “yes” or “potentially” to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project activity entails distribution of Greenway stoves for cooking among households and is not anticipated to have any impact on cultural heritage of the host country.

P.4.2 | FORCED EVICTION AND DISPLACEMENT

<u>P.4.2.1 </u>	Does the project involve any risks related to involuntary relocation of people?	☐ YES ☒ NO
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If the answer to question above is “yes,” please explain the reason and how the project will ensure compliance with applicable requirements.

The project activity entails distribution of Greenway stoves for cooking among households and does not entail imposing any form of eviction or displacement of people.

Would the project potentially involve or lead to:

ERROR! REFERENCE SOURCE NOT FOUND.	risk of forced evictions or involuntary relocation of people?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	temporary or permanent and full or partial physical displacement (including people without legally recognisable claims to land)?	☐ YES ☐ POTENTIALLY ☒ NO
<u>P.4.2.2 </u>	economic displacement (e.g., loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	☐ YES ☐ POTENTIALLY ☒ NO
<u>P.4.2.2 </u>	If answer to question above is “YES” or “POTENTIALLY”, - has the project developed Resettlement Action Plan or Livelihood Action Plan in consultation and agreement with affected individual, group or community? - has the project integrated Resettlement Action Plan or Livelihood Action Plan into the Project design?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to question above is “YES” – are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to question above is “YES”, have project design been changed, modified, updated considering opinions and recommendations of an Expert Stakeholder?	☐ YES ☐ NO ☒ NA

If the answer is “yes” or “potentially” to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project activity entails distribution of Greenway stoves for cooking among households and does not entail imposing any form of involuntary relocation, eviction or displacement of people.

P.4.3 | LAND TENURE AND OTHER RIGHTS

<u>P.4.3.1 </u>	Does the project involve any risks related to identifying and managing legitimate tenure rights that may be affected by the project?	☐ YES ☒ NO
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If the answer to question above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements.

The project activity entails distribution of Greenway stoves for cooking among households and does not involve land acquisition nor land use or tenure changes of the targeted community.

Would the project potentially involve or lead to:

ERROR! REFERENCE SOURCE NOT FOUND.	impacts on or changes to land tenure arrangements and/or community-based property rights/customary rights to land, territories and/or resources?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	uncertainties with regards to land tenure, access rights, usage rights or land ownership? Examples include, but are not limited to water access rights, community-based property rights and customary rights.	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Changes in legal arrangements, if yes, are the changes done in line with relevant laws and regulations?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	Changes in legal arrangements, if yes, are these changes agree with free, prior and informed consent of the involved stakeholders?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	Does some other entity (other than the project developer) hold uncontested land title for the entire Project Boundary?	☐ YES ☐ NO ☒ NA
<u>P.4.3.4 </u>	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to question above is "YES", have project design been changed, modified, updated considering opinions and recommendations of an Expert Stakeholder?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE	Have project developer in consultation with stakeholders established a functioning mechanism to receive, process, resolve, communicate and record grievances?	☐ YES ☐ NO ☒ NA

NOT FOUND.		
If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements. <i>The project activity entails distribution of Greenway stoves for cooking among households and does not involve change in land tenure, legal arrangements or user rights by the targeted community.</i>		
P.4.4 INDIGENOUS PEOPLES		
P.4.4.1 	Does the project involve Indigenous People within the Project area of influence who may be affected directly or indirectly by the Project?	☐ YES ☒ NO
If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements. <i>The project activity entails distribution of Greenway stoves for cooking among households and does not seek to influence or cause negative impact on any indigenous people who may be found within the project boundary.</i>		
Would the project potentially involve or lead to:		
ERROR! REFERENCE SOURCE NOT FOUND.	affect areas where indigenous peoples are present (including project area of influence)	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	affect areas, land and territory claimed by indigenous peoples?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.4.7 	If answer to above questions is "YES" or "POTENTIALLY", - Is it determined that the proposed project may affect the rights, lands, resources, or territories of indigenous people? - Has an "Indigenous People Plan" (IPP) or "Indigenous People Plan Framework" been elaborated and included in the project documentation? - Was the plan developed in accordance with the effective and meaningful participation of indigenous peoples and in accordance with UNDP Guidelines?	☐ YES ☐ NO ☒ NA
P.4.4.3 	risk of forcibly removing indigenous people from their lands and territories?	☐ YES ☐ POTENTIALLY

		☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	utilisation and/or commercial development of natural resources on lands and territories claimed by indigenous peoples? Consider, and where appropriate ensure, consistency with the answers under Principle 4.1 above	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to question above is "YES" or "POTENTIALLY" - Did the project obtain free, prior and informed consent from indigenous people before taking their cultural, intellectual, religious, and/or spiritual property? - Does the project ensure that the indigenous people receive an equitable sharing of benefits resulting from the use of their traditional knowledge and practices? - Does the project ensure that the sharing of benefits resulting from the use of indigenous peoples' traditional knowledge and practices is culturally appropriate and inclusive? - Does the project ensure that the provision of equitable sharing of benefits does not impede land rights or equal access to basic services including health services, clean water, energy, education, safe and decent working conditions, and housing?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project lack appropriate feedback and grievance channels for Indigenous Peoples and their representatives?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	Has a grievance mechanism not been established at the beginning of programme or project implementation with due consideration given to customary dispute settlement mechanisms among the Indigenous Peoples concerned and will it remain operational throughout the project cycle?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA
P.4.4.9	If answer to question above is "YES", have project design been changed, modified, updated considering opinions and recommendations of an Expert Stakeholder?	☐ YES ☐ NO ☒ NA

If the answer is “yes” or “potentially” to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project activity entails distribution of Greenway stoves for cooking among households and will negatively impact on areas, lands or territories occupied by indigenous people who may be found within the project boundary. On the contrary, using the project stoves saves on fuel used for cooking which has a positive impact in protecting forested areas which are important for indigenous communities.

P.5 | CORRUPTION

ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve, or is it complicit in, contributing to or reinforcing corruption or corrupt projects?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project have a risk of encouraging bribery, kickbacks, or other unethical behavior?	☐ YES ☒ NO

If the answer to any of the questions above is “yes,” please explain project situation and how the project will ensure compliance with applicable requirements.

The project developer and all the parties involved in implementation of the project will be guided by the values of integrity, transparency and accountability and will not engage in any corrupt practices or reinforce the same.

ECONOMIC SAFEGUARDING PRINCIPLES

ERROR! REFERENCE SOURCE NOT FOUND.**ERROR! REFERENCE SOURCE NOT FOUND.**

ERROR! REFERENCE SOURCE NOT FOUND.**ERROR! REFERENCE SOURCE NOT FOUND.**

ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve, facilitate, or condone forced labor, or pose a potential risk of forced labor?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project violate any labor or health and safety laws, international obligations, or ILO conventions?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project violate the principles of equal opportunity and fair treatment in its employment decisions?	☐ YES ☒ NO

<u>NOT FOUND.</u>		
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	Does the project violate national laws, if available regarding non-discrimination in employment?	☐ YES ☒ NO
<u>ERROR! REFERENCE SOURCE NOT FOUND. ERROR! REFERENCE SOURCE NOT FOUND.</u>	Does the project allow child labor?	☐ YES ☒ NO
<u>ERROR! REFERENCE SOURCE NOT FOUND. ERROR! REFERENCE SOURCE NOT FOUND.</u>	Does the project have insufficient processes and measures in place to ensure the safety and health of project workers?	☐ YES ☒ NO
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	Does the project have insufficient measures to safeguard and support vulnerable project workers, such as women, people with disabilities, migrant workers, and young workers, and to prevent any kind of harassment, abuse, bullying, or exploitation, including gender-based violence (GBV)?	☐ YES ☒ NO
<u>P.6.1.10 </u>	Does the project have no grievance mechanism available for workers to voice workplace concerns? Is information about this mechanism not provided to workers at the time of recruitment, or is it not easily accessible?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project implementer employs the employees on a voluntary basis without any form of coercion or forced labour. In addition, the project implementer only employs adults of legal age as per host country requirements. All the above activities do not involve use of child labour. The project implementer also operates within the provisions of the labour laws of India and respects the principles of equal employment, non-discrimination, employees' safety and right to information and raising concerns.

Would the project potentially involve or lead to:

(NOTE: APPLIES TO BOTH PROJECT AND CONTRACTOR WORKERS)

ERROR! REFERENCE SOURCE NOT FOUND.	use of forced labour?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	working conditions that do not meet national labour laws and international commitments?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	working conditions that may deny freedom of association and collective bargaining?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	absence of documented working agreements with all individual workers <i>if such agreements do not exist, or do not address working conditions and terms of employment, the project developer shall provide reasonable working conditions and terms of employment.</i>	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	use of migrant workers? <i>If engaged, the developer shall ensure that they are engaged substantially equivalent terms and conditions to non-migrant workers carrying out similar work.</i>	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	having no arrangements for basic services ²¹ for workers? <i>The project developer shall put in place and implement policies on the quality and management of the accommodation and provision of basic services in a manner consistent with the principles of non-discrimination and equal opportunity. Workers' accommodation arrangements should not restrict workers' freedom of movement or of association</i>	☐ YES ☐ POTENTIALLY ☒ NO
P.6.1.2 	any form of discrimination or harassment based on factors unrelated to job requirements, such as gender, race, nationality, ethnicity, social or indigenous origin, religion or belief, disability, age, or sexual orientation?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	any form of discrimination in any aspect of employment, such as recruitment, compensation, working conditions, training, job assignment, promotion, termination, or discipline?	☐ YES ☐ POTENTIALLY ☒ NO

²¹ Basic services requirements refer to minimum space, supply of water, adequate sewage and garbage disposal system, appropriate protection against heat, cold, damp, noise, fire, and disease-carrying animals, adequate sanitary and washing facilities, ventilation, cooking and storage facilities and natural and artificial lighting, and in some cases basic medical services.

ERROR! REFERENCE SOURCE NOT FOUND.	harassment, intimidation, and/or exploitation, especially in regard to women?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	discriminatory working conditions and/or lack of equal opportunity where national law provides provision to address non-discrimination in employment?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	use of child labour? (including third-party engaged workers)	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	inadequate and verifiable mechanisms for age verification?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	no processes and measures in place for the safety and health of project workers?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	No provision of safety and health training provisions, including on the proper use and maintenance of personal protective equipment conducted by competent persons and the maintenance of training records?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	No provision to record and document accidents, diseases, incidents, and any resulting injuries, illnesses, or deaths?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	No measures to protect vulnerable project workers from harassment, exploitation, and gender-based violence (GBV)? This includes women, people with disabilities, migrant workers, and young workers.	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	No grievance mechanism available for workers to voice workplace concerns.	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	No measures for due diligence and the establishment of policies and procedures to manage and monitor the performance of third-party employees in the project?	☐ YES ☒ NO

NOT FOUND.		
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If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The employees employed by the project have formal working arrangements as the employees' sign contracts outlining their roles and benefits. The contracts also outline their job descriptions. The working conditions follow laid down guidelines. All the health and safety related laws are abided by in the production and distribution process. The working conditions of the employees and workers are in compliance with national guidelines of the host country (India). As per occupational and safety guidelines, all workers using any equipment are trained on how to handle that equipment and any incident is documented. This will be monitored against principle P.3. Further, the project will not engage in any form of employee discrimination, intimidation, harassment, or exploitation during its implementation.

P.6.2 | NEGATIVE ECONOMIC CONSEQUENCES

ERROR! REFERENCE SOURCE NOT FOUND.	Is there a risk of project failure during implementation or after project certification due to a lack of financial resources?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project have potential negative impacts or pose a risk to the local economy?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Are there any potential risks or negative impacts this project may have on vulnerable or marginalised social groups, despite the benefits it may bring?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project distributes stoves which help end users reduce their fuel consumption. By reducing their fuel consumption, their associated expenses on fuel collection and purchases are reduced. Therefore, the project will not cause negative economic consequences during and after implementation. In addition, the project through carbon revenues sells the stove at a highly discounted price which helps end users to have better access to clean cooking an affordable price. The project will also create employment opportunities which will benefit the locals. These will have a positive impact to the local economy.

Would the project involve or lead to:

ERROR! REFERENCE SOURCE NOT FOUND.	economic impacts (negative/detrimental) to the local economy?	☐ YES ☐ POTENTIALLY ☒ NO
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ERROR! REFERENCE SOURCE NOT FOUND.	negative economic consequences during and after project implementation, e.g., for vulnerable and marginalised social groups in targeted communities?	☐ YES ☐ POTENTIALLY ☒ NO
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If the answer is “yes” or “potentially” to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The implementation of the project is anticipated to have net positive impact to the local economy through job creation and monetary savings which will benefit the targeted community including vulnerable and marginalised groups.

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P.7.1.1 	Does the project have a risk of increasing greenhouse gas emissions over the Baseline Scenario?	☐ YES ☒ NO
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If the answer to question above is “yes,” please explain project situation and how the project will ensure compliance with applicable requirements.

The project will deploy efficient cooking stoves against the traditional cooking stoves. Their use will ultimately result in GHG emissions reduction compared to the baseline scenario.

Would the project involve or lead to:

P.7.1.1 	increase greenhouse gas emissions over the Baseline Scenario?	☐ YES ☐ POTENTIALLY ☒ NO
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If the answer is “yes” or “potentially” to the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project will deploy efficient cooking stoves against the traditional cooking stoves. Their use will ultimately result in GHG emissions reduction compared to the baseline scenario.

P.7.2 |ENERGY SUPPLY

P.7.2.1 	Does the project pose a risk to the availability and reliability of energy supply to other users?	☐ YES ☒ NO
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If the answer to question above is “yes,” please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of improved cookstoves which will not impact on supply of energy to end-users or other people in the project area.

Would the project involve or lead to:

P.7.2.1	negative impact on the availability and reliability of energy supply to other users?	☐ YES ☐ POTENTIALLY ☒ NO
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If the answer is “yes” or “potentially” to the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project involves the dissemination of improved cookstoves which will not impact on supply of energy to end-users or other people in the project area.

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ERROR! REFERENCE SOURCE NOT FOUND.	Does the project increase water usage to a level that will not allow for the maintenance of environmental flows?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project result in the discharge of wastewater that does not meet the required standard for beneficial reuse and could therefore negatively impact the environmental flow?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project have the potential risk to exceed the rate of recharge for the groundwater source?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve any processes or activities that could contaminate the groundwater and render it unsuitable for use?	☐ YES ☒ NO

If the answer to any of the questions above is “yes,” please explain project situation and how the project will ensure compliance with applicable requirements.

The project entails distribution of cookstoves and doesn’t involve any activity related to discharge of wastewater, extraction of surface or ground water or activities that could contaminate water resources.

Would the project involve or lead to:

P.8.1.1	affect the natural or pre-existing pattern of watercourses, groundwater and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	☐ YES ☐ POTENTIALLY ☒ NO
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ERROR! REFERENCE SOURCE NOT FOUND.	Wastewater discharge of quality that does not meet the required standard for beneficial reuse?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	significant extraction, diversion of ground water? For example, construction of dams, reservoirs, river basin developments, groundwater extraction	☐ YES ☐ POTENTIALLY ☒ NO
P.8.1.2 	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA

If the answer is “yes” or “potentially” to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project entails distribution of cookstoves and doesn't involve any activity related to changing water courses, wastewater discharge or extraction and diversion of ground water.

P.8.2 | EROSION AND/OR WATER BODY INSTABILITY

P.8.2.1 	Does the project have a risk of negatively impacting the catchment and has it been assessed and addressed?	☐ YES ☒ NO
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If the answer to question above is “yes,” please explain project situation and how the project will ensure compliance with applicable requirements.

The project entails distribution of cookstoves and doesn't involve any that could negatively impact on water catchments.

Would the project involve or lead to:

P.8.2.2 - P.8.2.5 	negatively impact on the catchment area? <i>If yes, Erosion prevention measures, including soil and slope protection measures, must be implemented before project commencement. These measures should involve natural terracing, infiltration strips, permanent ground cover, hedge and tree rows, and effective slope length assessment. Regular reassessment of these measures is necessary.</i>	☐ YES ☐ POTENTIALLY ☒ NO
P.8.2.6 	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA

If the answer is “yes” or “potentially” to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project entails distribution of cookstoves and doesn't involve any that could negatively impact on water catchment areas.

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P.9.1.1 P.9.1.3 	Is there any risk of soil resource degradation or loss of ecosystem services provided by soils in the project? <i>If yes, the project shall maintain healthy soils by minimising negative impacts on soil health, productivity, structure, and water retention. Steps to minimise soil degradation include crop rotation, composting, using N-fixing plants, and reducing tillage and ecologically harmful substances.</i>	☐ YES ☒ NO
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If the answer to question above is “yes,” please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves, and the activities does not pose any immediate risk of soil degradation or loss of ecosystem services.

Would the project involve or lead to:

ERROR! REFERENCE SOURCE NOT FOUND.	production, harvesting, and/or management of living natural resources by small-scale landholders and/or local communities?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	if answer to above question “yes” or “potentially”, does project adopt appropriate and culturally sensitive sustainable resource management practices?	☐ YES ☐ NO ☒ NA

If the answer is “yes” or “potentially” to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project involves the dissemination of cookstoves, and the activities does entail harvesting of natural resources by small-scale holders or local communities.

P.9.2 | VULNERABILITY TO NATURAL DISASTER

P.9.2.1 	Does the project have any risks associated with natural or man-made hazards that could result from land use changes due to the project?	☐ YES ☐ NO*
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If the answer to question above is “yes,” please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves, and the activities does not pose a risk of imposing or increasing vulnerability to natural disasters.

Would the project involve or lead to:

ERROR! REFERENCE SOURCE NOT FOUND.	any potential risks that require emergency preparedness and response planning?	☐ YES ☐ POTENTIALLY ☐ NO*
ERROR! REFERENCE SOURCE NOT FOUND.	if answer to above question "yes" or "potentially", did the project developer disclose appropriate information about emergency preparedness and response to affected communities?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project involves the dissemination of cookstoves, and the activities does not introduce any risks that require emergency preparedness and response planning.

ERROR! REFERENCE SOURCE NOT FOUND.ERROR! REFERENCE SOURCE NOT FOUND.

ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve the transfer, handling, and use of genetically modified organisms/living modified organisms that may result in adverse effects on biological diversity?	☐ YES ☐ NO*
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If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves, and the activities does entail handling or use of GMOs.

Would the project involve or lead to:

ERROR! REFERENCE SOURCE NOT FOUND.	the transfer, handling and use of genetically modified organisms/living modified organisms (GMOs/LMOs) that result from modern biotechnology	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to above question is "yes" has a risk assessment by a competent Expert stakeholder been carried out in accordance with Annex iii of the Cartagena protocol on biosafety to the convention on biological diversity?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to above question is "yes" has any risks identified in the risk assessment?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	Forestry (for example Afforestation/Reforestation) involving GMO planting?	☐ YES ☐ NO ☒ NA

	<i>Note – Forestry projects (for example Afforestation/ Reforestation) involving GMO planting are not eligible for Certification under Gold Standard for the Global Goals.</i>	
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If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project involves the dissemination of cookstoves, and the activities does entail handling or use of GMOs. The project involves the dissemination of cookstoves, and the activities does entail handling or use of GMOs, or use of biotechnology.

P.9.4 | RELEASE OF POLLUTANTS

<u>P.9.4.1 </u>	Does the project have a risk of releasing pollutants to air, water, and land in routine, non-routine, or accidental circumstances?	☐ YES ☐ NO*
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If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves and will not result in release of pollutants during its implementation.

Would the project involve or lead to:

ERROR! REFERENCE SOURCE NOT FOUND.	any potential risk of pollutant release that cannot be avoided?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to above question is "Yes" or "potentially", has the project identified all potential pollution sources that may degrade the quality of soil, air, surface, and groundwater in the project area?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to above question is "Yes" or "potentially", do the pollution prevention and control technologies and practices applied during the project life cycle align with national regulations or international best practices?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to above question is "Yes", is there a monitoring plan to ensure that mitigation measures are implemented, and resources are protected?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project involves the dissemination of cookstoves and will not result in release of pollutants during its implementation.

P.9.5 | HAZARDOUS AND NON-HAZARDOUS WASTE

ERROR! REFERENCE	Does the project involve the generation of waste materials (both hazardous and non-hazardous)?	☐ YES ☐ NO
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<u>SOURCE NOT FOUND.</u>		
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	Does the project involve risk of release of hazardous materials resulting from their production, transportation, handling, storage, or use?	☐ YES ☐ NO
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	Does the project involve the use of any chemicals or materials subject to international bans or phase-outs?	☐ YES ☐ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves and does not entail use of chemicals, nor pose a risk due to generation of hazardous or non-hazardous waste materials.

Would the project involve or lead to:

<u>P.9.5.1</u>	the generation and management of waste materials?	☐ YES ☐ POTENTIALLY ☒ NO
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	treatment, destruction, or disposal of waste material?	☐ YES ☒ NO ☐ NA
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	If answer to above question is "Yes", does the project involve an environmentally friendly method that includes appropriate control of emissions and residues resulting from the handling and processing of waste material?	☐ YES ☐ NO ☒ NA
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	risk of release of hazardous materials resulting from their production, transportation, handling, storage, or use?	☐ YES ☒ NO ☐ NA
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	If answer to above question is "yes", does project has measures in place to address health risks?	☐ YES ☐ NO ☒ NA
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	Involve manufacture, trade, and use of chemicals and hazardous materials subject to international bans or phase-outs due to their high toxicity to living organisms, environmental persistence, potential for bioaccumulation, or potential for depletion of the ozone layer	☐ YES ☐ POTENTIALLY ☒ NO

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project involves the dissemination of cookstoves and does not involve manufacture or use of chemicals, nor pose a risk due to generation or management of hazardous or non-hazardous waste materials.

P.9.6 | PESTICIDES & FERTILISERS

ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve the use of chemical pesticides?	☐ YES ☐ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve purchase, store, manufacture, trade or use products that fall in Classes IA (extremely hazardous) and IB (highly hazardous)	☐ YES ☐ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project use fertilisers, and if so, are measures being taken to minimise their use and nutrient losses to the environment?	☐ YES ☐ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves and does not entail use of fertilisers and pesticides.

Would the project involve or lead to:

ERROR! REFERENCE SOURCE NOT FOUND.	chemical pesticides use for pest management?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to question above is "yes" or "potentially", does project has documented Chemical Pesticides Policy in place?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	purchase, store, use, manufacture, or trade in Class II (moderately hazardous) pesticides?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to question above is "yes" or "potentially", does project has appropriate controls on manufacture, procurement, or distribution and/or use of these chemicals?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project involves the dissemination of cookstoves and does not entail use of chemical pesticides.

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ERROR! REFERENCE SOURCE NOT FOUND.	Does the project have a risk of unsustainable forest management, including timber harvesting?	☐ YES ☐ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project pose a risk of depleting biodiversity and ecosystem functionality in areas where improved forest management is undertaken?	☐ YES ☐ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project risk not meeting requirements for environment-friendly, socially beneficial, and economically viable plantations using native species whenever possible?	☐ YES ☐ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves which does not involve harvesting of forests. Instead, the project will reduce demand for forest products such firewood collection through use of energy efficient cookstoves.

P.9.8 | FOOD SECURITY

P.9.8.1 	Does the project involve the risk of negatively influencing access to and availability of food for people affected?	☐ YES ☐ NO
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If the answer to the question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves which will not impact on food access and availability.

Would the project involve or lead to:

P.9.8.1 	modification of the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	☐ YES ☐ POTENTIALLY ☐ NO
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If the answer is "yes" or "potentially" to the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project involves the dissemination of cookstoves which will not impact quantity or nutritional quality of food.

ERROR! REFERENCE SOURCE NOT FOUND. ERROR! REFERENCE SOURCE NOT FOUND.		
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve any risks to animal welfare? Animal welfare shall be ensured by providing access to water and food, appropriate environment, humane treatment, and staff training. Evidence of mistreatment will be treated as an immediate non-conformity.	☐ YES ☐ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve any potential risk of excessive or inadequate use of veterinary medicines?	☐ YES ☐ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project involve the risk of administering synthetic growth promoters, including hormones?	☐ YES ☐ NO
If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
<i>The project involves the dissemination of cookstoves and does not entail activities that pose a risk to animal welfare, nor use of veterinary medicine or growth hormones.</i>		
Would the project involve or lead to:		
ERROR! REFERENCE SOURCE NOT FOUND.	animal husbandry or harvesting of fish populations or other aquatic species? ²²	☐ YES ☒ NO ☐ NA
ERROR! REFERENCE SOURCE NOT FOUND.	limiting access for animals to basic needs like drinking water, adequate food, daylight, appropriate shelter etc.?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	inadequate measures to isolate sick animals and control the spread of disease, especially zoonotic diseases?	☐ YES ☒ NO ☐ NA
ERROR! REFERENCE SOURCE NOT FOUND.	inadequate low-stress methods, equipment, and facilities that facilitate calm animal movement.	☐ YES ☒ NO ☐ NA

²² 'Involve' means if the project mechanism and/or impact(s) are achieved via changing animal husbandry practices in some way.

NOT FOUND.		
ERROR! REFERENCE SOURCE NOT FOUND.	inadequate measures to ensure that animals are exposed to the least stress possible during transportation and slaughtering?	☐ YES ☒ NO ☐ NA
ERROR! REFERENCE SOURCE NOT FOUND.	inappropriate spacing per animal and stocking rates per land unit?	☐ YES ☒ NO ☐ NA
P.9.9.8	inadequate measures to address the specific needs of aquatic animals?	☐ YES ☒ NO ☐ NA
ERROR! REFERENCE SOURCE NOT FOUND. ERROR! REFERENCE SOURCE NOT FOUND.	primary production of living natural resources such as animal husbandry, aquaculture, and fisheries? If the answer is yes, implement industry-standard sustainable management practices in line with to one or more relevant and credible standards and utilise available technologies.	☐ YES ☒ NO ☐ NA

If the answer is "yes" or "potentially" to any of the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project involves the dissemination of cookstoves and does not entail animal husbandry.

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ERROR! REFERENCE SOURCE NOT FOUND.	Does the project have the risk of negatively impacting HCV areas and/or critical habitats?	☐ YES ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project in the project area or area of downstream impacts have risks to the following: native tree patches, individual native trees, freshwater resources (including rivers, lakes, swamps, temporary water bodies, and wells), habitats of rare, threatened, and endangered species, and biodiversity-enhancing areas?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves which does pose a risk to high conservation value areas and critical habitats.

Would the project involve or lead to:		
ERROR! REFERENCE SOURCE NOT FOUND.	identified habitats as HCV areas and or Critical habitats?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to above question is "yes", does the project have any risks that could negatively impact the catchment, project success, and surrounding HCV and ecological assets, as well as any measurable adverse impacts on the criteria or biodiversity values for which the critical habitat was designated, and on the ecological processes supporting that biodiversity?	☐ YES ☐ NO ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	If answer to above question is "yes", is a robust, appropriately designed, and long-term Habitats and Biodiversity Action Plan absent which will make the project unable to achieve net gains of those biodiversity values for which the critical habitat was designated?	☐ YES ☐ NO ☒ N/A
ERROR! REFERENCE SOURCE NOT FOUND.	Does the project area or area of downstream impacts have native tree patches, individual native trees, freshwater resources (including rivers, lakes, swamps, temporary water bodies, and wells), habitats of rare, threatened, and endangered species, and biodiversity-enhancing areas?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	If the answer to the above question is "yes", will the project have any adverse effects on these areas?	☐ YES ☐ No ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	If the answer to above question is "yes", does the project has opportunities to minimise unwarranted conversion or degradation of the habitat and to enhance the habitat as part of its development?	☐ YES ☐ No ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	Is the project applying Land Use & Forest Activity Requirements and managing a minimum 10% of the project area to protect or enhance the biological diversity of native ecosystems following HCV approach as per the given requirements?	☐ YES ☐ No ☒ NA
ERROR! REFERENCE SOURCE NOT FOUND.	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA
If the answer is "yes" or "potentially" to any of the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.		

The project involves the dissemination of cookstoves which does pose a risk to high conservation value areas and critical habitats.

P.9.11 | ENDANGERED SPECIES

<u>P.9.11.1 </u>	Does the project lead to the reduction or negative impact on any recognised Endangered, Vulnerable or Critically Endangered species?	☐ YES ☒ NO
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If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves, and it's not expected to negatively impact on any recognised Endangered, Vulnerable or Critically Endangered species within the targeted areas.

Would the project involve or lead to:

<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	distortion of habitats of endangered species?	☐ YES ☐ POTENTIALLY ☒ NA
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	If answer to the above question is "yes", does the project plan to protect and enhance them?	☐ YES ☐ POTENTIALLY ☐ NO ☒ N/A
<u>ERROR! REFERENCE SOURCE NOT FOUND.</u>	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

The project involves the dissemination of cookstoves, and it's not expected to negatively impact on any habitats of endangered species within the targeted areas.

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<u>P.9.12.1 </u>	Does project introduce any alien species (not currently established in the country or region of the project) into new environments?	☐ YES ☒ NO
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If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

The project involves the dissemination of cookstoves and does not entail introduction of invasive alien species.

Would the project involve or lead to:		
ERROR! REFERENCE SOURCE NOT FOUND.	risk of introducing any alien species with a high risk of invasive behaviour regardless of whether such introductions are permitted under the existing regulatory framework?	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	risk of potential accidental or unintended introductions including the transportation of substrates and vectors (such as soil, ballast, and plant materials) that may harbour alien species.	☐ YES ☐ POTENTIALLY ☒ NO
ERROR! REFERENCE SOURCE NOT FOUND.	risk of spreading alien species into areas in which they have not already been established?	☐ YES ☐ POTENTIALLY ☒ NO
If the answer is "yes" or "potentially" to any of the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.		
<i>The project involves the dissemination of cookstoves and does not entail introduction or spread of invasive alien species.</i>		

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organisation name	Greenway Grameen Infra Private Limited
Registration number with relevant authority	U70102MH2010PTC210549 under Companies Act, 1956, India
Street/P.O. Box	#501-502, Plot No.522, CTS No. F/355, Makani Centre, 35th road, T.P.S. III,
Building	Makani Centre
City	Bandra (W) Mumbai, Mumbai City
State/Region	Maharashtra
Postcode	400050
Country	India
Telephone	+91-22-60522638
E-mail	contact@greenwayappliances.com
Website	https://www.greenwayappliances.com/
Contact person	Mr. Antik Mathur
Title	CEO
Salutation	Mr.
Last name	Mathur
Middle name	
First name	Antik
Department	Management
Mobile	
Direct tel.	
Personal e-mail	

Organisation name	SDG 13 VENTURES PTE. LTD
Registration number with relevant authority	201939197D
Street/P.O. Box	100 Peck Seah Street, #10-19, PS100
Building	Singapore
City	
State/Region	
Postcode	079333
Country	Singapore
Telephone	+65 69620655
E-mail	
Website	Impact Investment SDG 13 Ventures
Contact person	Sonal Kohli
Title	President, Operations
Salutation	Mrs.
Last name	Kohli
Middle name	
First name	Sonal
Department	Management
Mobile	
Direct tel.	
Personal e-mail	sonal.k@sdg13ventures.com

Revision History

Version	Date	Remarks
2.3	Dd/mm/yyyy	Editorial changes in line with V2.1 of the Safeguarding Principles and Requirements
2.2	21 June 2023	Editorial changes in line with V2.0 of the Safeguarding Principles and Requirements
2.1	14 April 2023	Integrated the design change memo as annex of the document.
2.0	4 May 2022	
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption